Lecture Notes: Lectures 1–10

Notes: Suhan Woo

The arc of the course

In the first lecture the course was laid out in three stages: first *modelling* dynamical systems — by transfer functions and in state-variable form — and learning to *analyse* the models; then a middle stretch that leans on linear algebra and on MATLAB; and finally stability, controller design, and classical control. The destinations named were PID control, state-feedback controllers, state observers, tracking controllers, and Bode and Nyquist plots. The Parts of these notes follow that plan.

- Part I — Feedback and Modelling §§1–2
- Part II — State-Space Models and Linearization §§3–4
- Part III — Transfer Functions §5
- Part IV — Frequency Response and Stability §§6–7
- Part V — Linear Algebra for State Space §§8–11, in progress
- *Stability, controller design, and classical control* — state-feedback controllers, state observers, tracking controllers, PID control planned

The planned items are listed as the first lecture described them; how they divide into Parts will be settled as the lectures arrive. The goal of Question 2.1 — bringing the two carts to rest 1 m ahead after 5 s without knowing their parameters — is a controller-design problem; the Parts so far build the models and the analysis that design will need.

How these notes are organised

These notes are ordered by topic rather than by date. They are divided into Parts, each a stage of the course, and every new lecture is folded into the Part and section it belongs to, so nothing is said twice. Section numbers run on through the whole document; they do not restart with each Part. A hairline rule carrying a lecture number and date marks the point where each lecture's new material begins, and the table below lists the same points; where several lectures are written up together as one unit, a single rule carries the whole range. **Remarks** and worked **Examples** fill in derivations, examples and figures beyond what fitted on the board, and sit with the topic they belong to — so a rule marks where a lecture began, not a claim that everything after it was said in that lecture.

Lecture	Date	Begins at
1	Wed, 2 Sep 2026	§1
2	Mon, 7 Sep 2026	§2
3	Wed, 9 Sep 2026	§4
4	Mon, 14 Sep 2026	§5
5	Wed, 16 Sep 2026	§5.6
6	Mon, 21 Sep 2026	§6.6
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Part I — Feedback and Modelling

Lecture 1 · 2 September 2026

1 What Control Engineering Is About

1.1 The question

Control engineering asks one question:

how should the input be chosen so that the output does what we want?

To answer it we must first know how the input and the output are related, and for a physical system that relation is almost always a *differential equation*. So the course begins with differential equations, not with controllers.

Two words will follow us all semester. The first is **uncertainty**: the differential equation we write down is never exactly the one the world obeys. The second is **feedback**: measuring the output and feeding it back into the input is what makes a design survive that uncertainty.

A control system in practice is assembled from sensors, actuators, computers, the controller itself, and the communication between them. In these notes all of that is idealised away into a signal $u(t)$ going in and a signal $y(t)$ coming out. The *output* y need not be a continuously varying quantity to be tracked; it may equally well be "get to that point and stay there". Both are handled by the same machinery.

1.2 Dynamic versus static

A *static* relation determines y at time t from u at the same time t. A *dynamic* relation has memory: the present output depends on the past input. The simplest example is a first-order linear differential equation,

$$L\dot{y} + Ry = u(t), \tag{1}$$

which is, for instance, an RL circuit driven by a voltage source u, with y the current. We write $y^*(t)$ for the *desired* output — the thing the controller is supposed to produce — and $u(t)$ for the control input.

Note 1.1. The symbol y is used for whatever the output happens to be: a current in (1), a position in the next example. Keep track of which is which; the mathematics does not care, but the units do.

1.3 A first example: holding a mass up against gravity

Consider a mass m that can move vertically, with position $y(t)$ measured upwards, acted on by a control force $u(t)$ and by gravity (Figure 1):

$$m\ddot{y} = u - mg. \tag{2}$$

We want to hold it at height h: the desired output is the constant $y^*(t) = h$, and the mass starts there at rest, $y(0) = h$, $\dot{y}(0) = 0$.

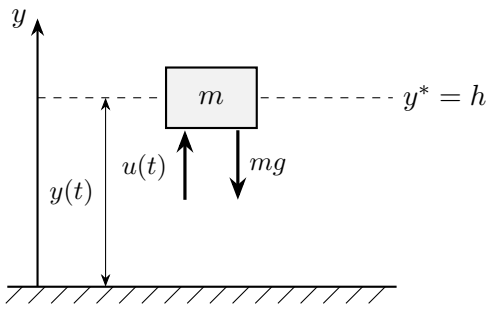


Figure 1. A mass held against gravity by a control force. The control problem is to choose $u(t)$ so that $y(t)$ stays at h.

(a) Open loop. Setting $\ddot{y} = 0$ in (2) gives $u = mg$. With that constant input $\ddot{y} \equiv 0$, and with the initial conditions above $y(t) \equiv h$ for all time. On paper, the problem is solved without any feedback at all.

(b) The trouble: uncertainty. We do not know m. We know an estimate $\hat{m}$, and so we apply $u = \hat{m}g$. Now (2) reads $m\ddot{y} = \hat{m}g - mg = (\hat{m} - m)g$, a *constant nonzero acceleration*. Integrating twice,

$$y(t) = h + \frac{(\hat{m} - m)g}{2m}\, t^2 \;\longrightarrow\; \pm\infty.$$

However small the modelling error $\hat{m} - m$, the mass eventually leaves. The open-loop design is not merely inaccurate; it fails qualitatively.

(c) The cure: feedback. Measure y and $\dot{y}$ and use

$$u(t) = -c\,\dot{y} - k\,(y - h) + \hat{m}g, \tag{3}$$

where c and k are *gains*: numbers we choose. They carry no uncertainty, because we are the ones who pick them. Substituting (3) into (2),

$$m\ddot{y} + c\,\dot{y} + k\,(y - h) = (\hat{m} - m)g. \tag{4}$$

For $c, k > 0$ this is a damped second-order system driven by a constant, so y settles, and setting $\ddot{y} = \dot{y} = 0$ gives the resting value

$$y(\infty) = h + \frac{(\hat{m} - m)g}{k}. \tag{5}$$

The error is now *bounded*, and it shrinks as k grows. Feedback did not remove the uncertainty; it converted an unbounded error into a small one.

Question 1.2. Why does this work? What are good values of c and k? How large may k be before something else goes wrong? Answering these is what the rest of the course is for.

1.4 Uncertainty, robustness, and learning

Designing a controller for the nominal model is only half the job; the other half is guaranteeing that it still behaves over the whole *set* of plants that the real system might be. That is the subject of *robust* control engineering.

As an aside, the lecture contrasted this with reinforcement learning, where a policy is trained from a reward signal rather than from a model. Its characteristic difficulty is the *sim-to-real gap*: a policy tuned in simulation need not work on hardware. The standard remedies — domain randomisation and friends, i.e. training over a randomised family of simulators so that no single mismatch is fatal — are robustness ideas in disguise. They are heuristic, but they are valuable exactly where a usable physical model is hard to come by. A model-based design, by contrast, is mathematical: one can in principle reason backwards from a specification to a controller. Even that is not easy, and much of this course is about learning how.

Lecture 2 · 7 September 2026

2 Modelling: the Two-Cart System

2.1 The running example

The example that will accompany us through the course is the *two-cart system* of Figure 2: a cart of mass m_1 at position $p_1(t)$, pushed by a force $u(t)$, coupled through a spring of stiffness k and a damper b to a second cart of mass m_2 at position $p_2(t)$. Both carts feel ground friction with coefficient c. The measured output is the position of the second cart, $y(t) = p_2(t)$. We take $p_1(0) = p_2(0) = 0$: positions are measured from where the carts start.

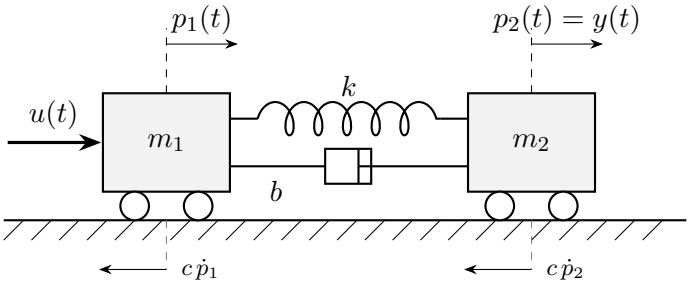


Figure 2. The two-cart system. The spring k and the damper b act between the carts; the ground friction c acts on each cart separately and opposes its velocity.

Question 2.1 (the goal for the semester)**.** Make the system move forward and *come to rest* $1\,m$ *ahead after* 5 *seconds* — and do it without knowing the values of m_1, k, m_2.

The second half of Question 2.1 is the real content. Anyone can hit a target when the plant is known exactly; a controller that works without that knowledge is a control design.

2.2 Equations of motion

Newton's law for each cart. Cart 1 is pushed by u, pulled by the spring towards cart 2, damped by the relative motion of the two carts, and dragged by ground friction; cart 2 sees the same, minus the input:

$$\begin{cases} m_1\ddot{p}_1 = u(t) + k\,(p_2 - p_1) + b\,(\dot{p}_2 - \dot{p}_1) - c\,\dot{p}_1, \\ m_2\ddot{p}_2 = \qquad\quad k\,(p_1 - p_2) + b\,(\dot{p}_1 - \dot{p}_2) - c\,\dot{p}_2, \\ \quad y = p_2. \end{cases} \tag{6}$$

Remark 2.2 (the sign of the coupling terms). A damper mounted in parallel with the spring resists the *relative* motion of its two ends, so its force on cart 1 has exactly the same sign pattern as the spring force: $+\,k(p_2 - p_1)$ and $+\,b(\dot{p}_2 - \dot{p}_1)$, and the mirror images on cart 2. A quick way to see that these signs are the physical ones is to look at the stretch of the spring, $q = p_1 - p_2$. Subtracting the two equations (with $c = 0$ for simplicity),

$$\ddot{q} + b\Big(\tfrac{1}{m_1} + \tfrac{1}{m_2}\Big)\dot{q} + k\Big(\tfrac{1}{m_1} + \tfrac{1}{m_2}\Big)q = \frac{u}{m_1},$$

a damped oscillator: the internal mode, in which the carts swing against each other, loses energy through the damper, as it should. Flipping the sign of either coupling term would give that mode negative stiffness or negative damping.

Note 2.3 (symbols). Throughout, c is the *ground friction* coefficient and b is the *damper between the carts*.

2.3 Three ways to describe the same system

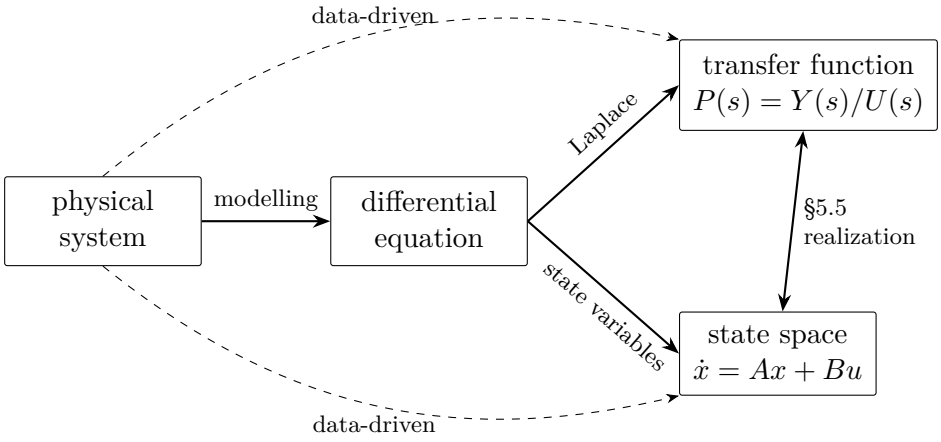


Figure 3. Routes from a physical system to a mathematical model. The solid path is this course. The dashed ones are system identification / learning from data (neural networks and the like), which skip the differential equation altogether.

The transfer function and the state-space form are two descriptions of the same differential equation, and (for linear systems) one can pass between them — the double arrow in Figure 3. Section 5.5 does one of the two directions explicitly.

Part II — State-Space Models and Linearization

3 The State-Space Representation

3.1 State variables

The equations (6) are second order. Introduce one variable per independent piece of "memory" the system has — here, a position and a velocity per cart:

$$x_1 = p_1, \qquad x_2 = \dot{p}_1, \qquad x_3 = p_2, \qquad x_4 = \dot{p}_2.$$

Then (6) becomes four *first-order* equations:

$$\begin{cases} \dot{x}_1 = x_2, \\ \dot{x}_2 = \dfrac{k}{m_1}(x_3 - x_1) - \dfrac{c}{m_1}x_2 + \dfrac{b}{m_1}(x_4 - x_2) + \dfrac{1}{m_1}u, \\ \dot{x}_3 = x_4, \\ \dot{x}_4 = \dfrac{k}{m_2}(x_1 - x_3) - \dfrac{c}{m_2}x_4 + \dfrac{b}{m_2}(x_2 - x_4), \\ y = x_3. \end{cases} \tag{7}$$

The four equations are thoroughly tangled together, but every one of them is first order, and that is precisely what makes them easy to handle — by a computer and by linear algebra alike.

3.2 The standard form

Collecting $x = (x_1, x_2, x_3, x_4)^{\mathsf{T}}$ we reach the **standard form of the state-space representation**:

$$\boxed{\dot{x}(t) = A\,x(t) + B\,u(t), \qquad y(t) = C\,x(t) + D\,u(t).} \tag{8}$$

Its dimensions are

$$x(t) \in \mathbb{R}^n, \qquad u(t) \in \mathbb{R}^m, \qquad y(t) \in \mathbb{R}^p, \tag{9}$$

so that $A \in \mathbb{R}^{n\times n}$, $B \in \mathbb{R}^{n\times m}$, $C \in \mathbb{R}^{p\times n}$, $D \in \mathbb{R}^{p\times m}$. The space $\mathbb{R}^n$ is called the **state space**, and n — the number of state variables — is the *order* of the system. A system with $m = p = 1$ is called SISO (single input, single output); otherwise it is MIMO. Note that the three dimensions are independent: a plant may have many states and a single input, or several inputs and one output.

For the two-cart system $n = 4$, $m = p = 1$. Taking $b = 0$ to keep the entries short,

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -\dfrac{k}{m_1} & -\dfrac{c}{m_1} & \dfrac{k}{m_1} & 0 \\ 0 & 0 & 0 & 1 \\ \dfrac{k}{m_2} & 0 & -\dfrac{k}{m_2} & -\dfrac{c}{m_2} \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ \dfrac{1}{m_1} \\ 0 \\ 0 \end{bmatrix}, \quad C = \begin{bmatrix} 0 & 0 & 1 & 0 \end{bmatrix}, \quad D = 0. \tag{10}$$

Remark 3.1 (where the uncertainty lives). Rows 1 and 3 of A contain no parameters at all. They are not physics; they are the definitions $\dot{x}_1 = x_2$ and $\dot{x}_3 = x_4$, and they are exactly right whatever the carts weigh. All the uncertainty of the model is confined to rows 2 and 4. It is a general and useful feature of mechanical state-space models: the kinematic rows are exact, and only the rows that encode Newton's law carry the parameters one might not know.

Remark 3.2 (keeping the damper). If b is not dropped, only rows 2 and 4 change:

$$A_{2\bullet} = \Big[-\tfrac{k}{m_1},\ -\tfrac{c+b}{m_1},\ \tfrac{k}{m_1},\ \tfrac{b}{m_1}\Big], \qquad A_{4\bullet} = \Big[\tfrac{k}{m_2},\ \tfrac{b}{m_2},\ -\tfrac{k}{m_2},\ -\tfrac{c+b}{m_2}\Big].$$

The damper adds to the diagonal velocity terms and couples the two velocities, exactly as the spring couples the two positions.

In MATLAB the quadruple is packaged as `sys = ss(A,B,C,D)`.

3.3 What "state" means

Definition 3.3 (state)**.** The *state* $x(t)$ is the minimum amount of information about the system at time t which, together with the input $u(\tau)$ for $\tau \geq t$, determines the output $y(\tau)$ for all $\tau \geq t$.

Everything that happened before time t matters only through $x(t)$. For the two carts, knowing where both carts are and how fast they are going is enough; their history is irrelevant. The word *minimum* is what makes the definition useful: it is why the state is four-dimensional here and not five.

Remark 3.4. Definition 3.3 says nothing about linearity, and indeed the state-space description works for nonlinear systems just as well (Section 4). The transfer function does not: it exists only for linear systems.

3.4 Simulating the standard form

Because (8) is a system of first-order equations, a computer can march it forward directly. The crudest such scheme, and the one this course starts with, is the *forward Euler method*: for a small step $h > 0$,

$$x(t + h) \ \approx \ x(t) + h\,\big(Ax(t) + Bu(t)\big), \tag{11}$$

or, for a general nonlinear right-hand side, $x(t + h) \approx x(t) + h\, f(x(t), u(t))$. It is nothing more than the first two terms of the Taylor series of x about t, so the error made in one step is of order h^2. Choosing h is not free, however: too large a step and the simulation diverges even though the system does not.

Remark 3.5 (how large a step). The limit on h can be computed exactly on the scalar test equation $\dot{x} = \lambda x$, whose true solution decays whenever $\operatorname{Re}\lambda < 0$. One Euler step multiplies x by a fixed factor,

$$x_{k+1} = (1 + h\lambda)\, x_k, \qquad x_k = (1 + h\lambda)^k x_0,$$

so the simulation decays only if $|1 + h\lambda| < 1$ — that is, only if $h\lambda$ lies inside the unit disc centred at -1. For a real $\lambda < 0$ this means

$$h < \frac{2}{|\lambda|}.$$

For the linear system $\dot{x} = Ax$ the same condition must hold for every eigenvalue of A at once, so the fastest mode sets the step. An eigenvalue on or near the imaginary axis — a lightly damped oscillation — is the worst case: for $\lambda = -\sigma \pm j\omega$ with small σ the condition $|1 + h\lambda| < 1$ reads $h < 2\sigma/(\sigma^2 + \omega^2)$, which is tiny when $\sigma \ll \omega$. A purely imaginary λ can never satisfy it: forward Euler turns every undamped oscillation into a growing one.

Lecture 3 · 9 September 2026

4 Nonlinear Systems, Equilibria, and Linearization

4.1 The nonlinear standard form

Most real systems are not linear. The general form we will use is

$$\dot{z} = f(z, u), \qquad w = h(z, u), \tag{12}$$

with $z(t) \in \mathbb{R}^n$, $u(t) \in \mathbb{R}^m$, $w(t) \in \mathbb{R}^p$ and

$$f : \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}^n, \qquad h : \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}^p$$

assumed *regular* — smooth enough that the derivatives below exist. Written out, (12) is

$$\dot{z} = \begin{pmatrix} \dot{z}_1 \\ \vdots \\ \dot{z}_n \end{pmatrix} = \begin{pmatrix} f_1(z(t), u(t)) \\ \vdots \\ f_n(z(t), u(t)) \end{pmatrix}.$$

Note 4.1 (notation for this section). We reserve (z, u, w) for the nonlinear system and (x, u, y) for the linear one, because the variables of the linearized model will turn out to be *deviations* from an operating point, not the physical quantities themselves. Mixing the two up is the single most common error in this material.

Example 4.2 (pendulum)**.** A pendulum of mass m on a massless rod of length ℓ, with viscous friction b at the pivot and a torque $u(t)$ applied there (Figure 4), obeys

$$m\ell^2\ddot{\theta} = u(t) - mg\ell\sin\theta - b\,\dot{\theta}, \qquad \text{i.e.} \qquad m\ell^2\ddot{\theta} + b\,\dot{\theta} + mg\ell\sin\theta = u(t),$$

where θ is measured from the downward vertical. With $z_1 = \theta$, $z_2 = \dot{\theta}$ this is exactly of the form (12):

$$\begin{cases} \dot{z}_1 = z_2, \\ \dot{z}_2 = -\dfrac{b}{m\ell^2}\, z_2 - \dfrac{g}{\ell}\sin z_1 + \dfrac{1}{m\ell^2}\, u, \\ w = z_1. \end{cases} \tag{13}$$

The $\sin z_1$ is the whole difficulty: everything else is linear.

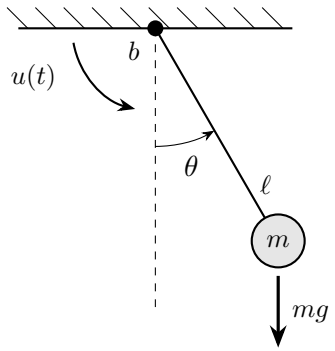


Figure 4. The pendulum of Example 4.2: θ is measured from the downward vertical, b is viscous friction at the pivot, and $u(t)$ is an applied torque.

4.2 Equilibria

The trick for dealing with (12) is to work near a point where nothing is happening.

Definition 4.3 (equilibrium)**.** A pair (z^*, u^*) with $f(z^*, u^*) = 0$ is an *equilibrium*. If the input is held at the constant value u^* and the system starts at $z(0) = z^*$, then $\dot{z} = 0$ and $z(t) \equiv z^*$ forever.

Note that the input must be a *constant* for this to make sense; "an equilibrium" is always an equilibrium *of a particular constant input*.

For the pendulum with $u^* = 0$: $f = 0$ forces $z_2^* = 0$ and $\sin z_1^* = 0$, so $z_1^* = 0$ (hanging down) or $z_1^* = \pi$ (balanced upright). Both are equilibria; only one of them is one you can leave unattended.

4.3 The Jacobian

For $f : \mathbb{R}^n \to \mathbb{R}^n$ (suppressing u for a moment), the *Jacobian* is the matrix of all first partial derivatives,

$$\mathrm{D}f \;=\; \begin{pmatrix} \dfrac{\partial f_1}{\partial x_1} & \dfrac{\partial f_1}{\partial x_2} & \cdots & \dfrac{\partial f_1}{\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \dfrac{\partial f_n}{\partial x_1} & \dfrac{\partial f_n}{\partial x_2} & \cdots & \dfrac{\partial f_n}{\partial x_n} \end{pmatrix} \;\in\; \mathbb{R}^{n\times n}.$$

It is the best linear approximation of f near a point — the multivariable analogue of the derivative.

4.4 Linearization

Expand f in a Taylor series about the equilibrium (z^*, u^*):

$$\dot{z} = f(z,u) = \underbrace{f(z^*,u^*)}_{=0} + \underbrace{\left.\frac{\partial f}{\partial z}\right|_{z^*,u^*}}_{=:A}\underbrace{(z-z^*)}_{=:x} + \underbrace{\left.\frac{\partial f}{\partial u}\right|_{z^*,u^*}}_{=:B}\underbrace{(u-u^*)}_{=:\tilde{u}} + \cdots$$

The constant term vanishes precisely because we expanded about an equilibrium — that is what equilibria are for. Since z^* is constant, $\dot{x} = \dot{z}$, and dropping the higher-order terms leaves

$$\dot{x} = A\,x + B\,\tilde{u}, \qquad A = \mathrm{D}_z f\big|_{z^*,u^*}, \qquad B = \mathrm{D}_u f\big|_{z^*,u^*}. \tag{14}$$

The output is treated the same way. From $w = h(z,u) = h(z^*,u^*) + \mathrm{D}_z h|_*\, x + \mathrm{D}_u h|_*\, \tilde{u} + \cdots$, define the output deviation $y := w - h(z^*,u^*)$, and then

$$y = C\,x + D\,\tilde{u}, \qquad C = \mathrm{D}_z h\big|_{z^*,u^*}, \qquad D = \mathrm{D}_u h\big|_{z^*,u^*}. \tag{15}$$

The result is exactly the standard form (8), which is why everything built for linear systems can be pointed at a nonlinear one — locally.

> **Warning (what the variables mean).** A, B, C, D are *constant* matrices: the Jacobians evaluated at the equilibrium, not functions of z. And x, $\tilde{u}$, y are all *deviations* — $x = z - z^*$, $\tilde{u} = u - u^*$, $y = w - h(z^*,u^*)$. (It is common to drop the tilde and write u for the input deviation as well.) If you feed a linearized model the absolute input instead of the deviation, everything downstream is wrong by a constant.

Example 4.4 (pendulum, continued)**.** For Example 4.2,

$$\mathrm{D}_z f = \begin{bmatrix} 0 & 1 \\ -\dfrac{g}{\ell}\cos z_1 & -\dfrac{b}{m\ell^2} \end{bmatrix}, \qquad \mathrm{D}_u f = \begin{bmatrix} 0 \\ \dfrac{1}{m\ell^2} \end{bmatrix}, \qquad x = \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} - z^*.$$

At the downward equilibrium $z^* = (0,0)$, $u^* = 0$, we get $\cos z_1^* = 1$ and

$$A = \begin{bmatrix} 0 & 1 \\ -\dfrac{g}{\ell} & -\dfrac{b}{m\ell^2} \end{bmatrix}, \qquad B = \begin{bmatrix} 0 \\ \dfrac{1}{m\ell^2} \end{bmatrix}, \qquad C = \begin{bmatrix} 1 & 0 \end{bmatrix}, \qquad D = 0.$$

Example 4.5 (pendulum, the upright equilibrium)**.** The $(2,1)$ entry of $\mathrm{D}_z f$ is $-\frac{g}{\ell}\cos z_1^*$, so it depends on where we linearize. At the upright equilibrium $z^* = (\pi, 0)$, $u^* = 0$ it becomes $+g/\ell$:

$$A_{\mathrm{up}} = \begin{bmatrix} 0 & 1 \\ \dfrac{g}{\ell} & -\dfrac{b}{m\ell^2} \end{bmatrix}, \qquad \det(\lambda I - A_{\mathrm{up}}) = \lambda^2 + \frac{b}{m\ell^2}\lambda - \frac{g}{\ell}.$$

The constant term of the characteristic polynomial is now negative, so one root is positive whatever b is — for $b = 0$ the eigenvalues are $\pm\sqrt{g/\ell}$. The linearization is unstable, as it must be: a pendulum balanced on its end falls over. At the hanging equilibrium the same polynomial reads $\lambda^2 + \frac{b}{m\ell^2}\lambda + \frac{g}{\ell}$, whose roots have negative real part for every $b > 0$. One nonlinear model, two equilibria, two qualitatively different linear models — which is exactly why an equilibrium must always be named when a linearization is written down.

4.5 Where the linear model sits

The linearized system is not a separate object; it is a window onto the real one. Figure 5 shows the bookkeeping: shift the input by u^* on the way in, and subtract the equilibrium output $h(z^*,u^*)$ on the way out.

4.6 Phase portraits

For a second-order autonomous system $\dot{x}_1 = f_1(x_1,x_2)$, $\dot{x}_2 = f_2(x_1,x_2)$, one can draw the whole solution family at once: plot the trajectories in the (x_1,x_2) plane. The picture is called a *phase portrait*. Time itself does not appear on it, but everything else does — equilibria are the points

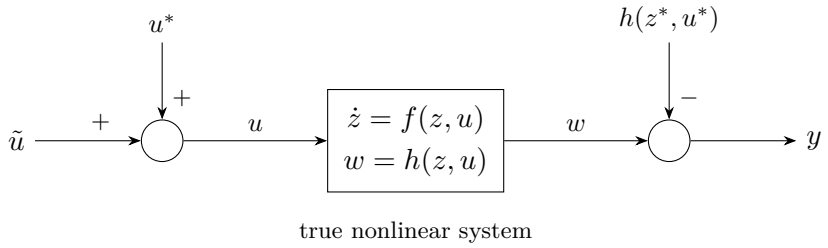


Figure 5. The linearized model $\dot{x} = Ax + B\tilde{u}$, $y = Cx + D\tilde{u}$ approximates the map $\tilde{u} \mapsto y$ drawn here, for $\tilde{u}$ small.

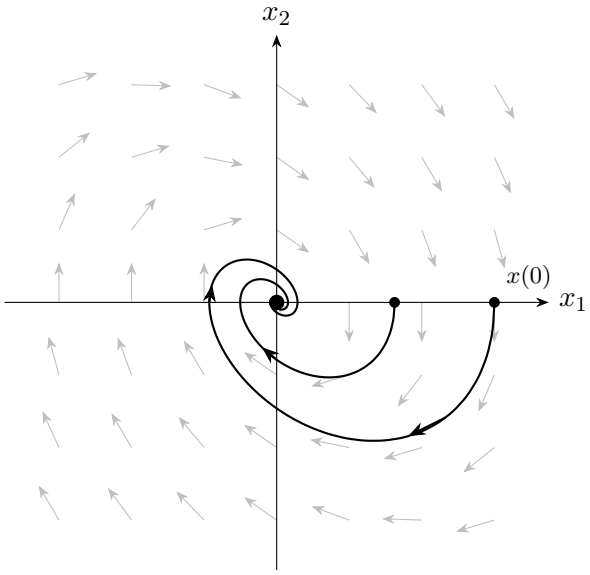


Figure 6. Phase portrait of $\dot{x}_1 = x_2$, $\dot{x}_2 = -x_1 - 0.7x_2$. Grey arrows are the vector field; the two heavy curves are trajectories. The black dot at the origin is the equilibrium, and both trajectories spiral into it.

where the vector field vanishes, and by uniqueness of solutions two trajectories never cross. Figure 6 shows a stable spiral.

For a *linear* second-order system $\dot{x} = Ax$ the whole portrait is fixed by the two eigenvalues $\lambda_{1,2}$ of A, and there are only a handful of possible pictures:

eigenvalues	portrait	behaviour near the equilibrium
real, both negative	stable node	every trajectory enters, without circling
real, both positive	unstable node	every trajectory leaves
real, opposite signs	saddle	in along one direction, out along the other
complex, $\operatorname{Re}\lambda < 0$	stable focus (spiral)	trajectories spiral in
complex, $\operatorname{Re}\lambda > 0$	unstable focus	trajectories spiral out
purely imaginary	centre	closed orbits, neither in nor out

For real eigenvalues the straight-line trajectories along the eigenvectors are the skeleton of the picture: the solution is $x(t) = c_1 e^{\lambda_1 t} v_1 + c_2 e^{\lambda_2 t} v_2$, and each component grows or decays along its

own eigenvector (Figure 7). For a complex pair $\sigma \pm j\omega$ the solution is an $e^{\sigma t}$ envelope around a rotation at frequency ω — exactly the partial-fraction terms of §5.7.

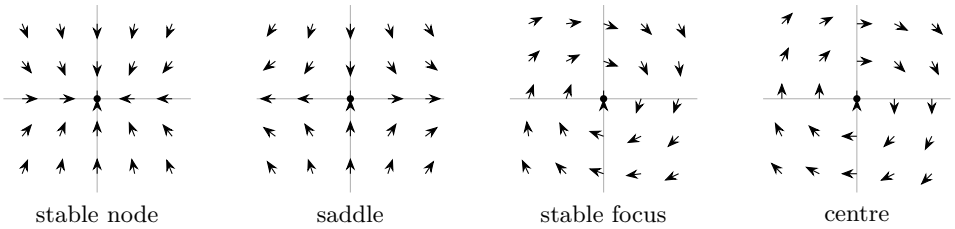


Figure 7. Four of the linear portraits of the table, drawn as vector fields of $\dot{x} = Ax$: a stable node ($A = \mathrm{diag}(-1, -2.5)$), a saddle ($\mathrm{diag}(1, -1.5)$), a stable focus (eigenvalues $-0.35 \pm j$), and a centre (eigenvalues $\pm j$).

Remark 4.6 (back to the pendulum). Linearization tells us which picture a nonlinear system shows *near* each equilibrium. For the pendulum of Example 4.2, the hanging equilibrium has characteristic polynomial $\lambda^2 + \frac{b}{m\ell^2}\lambda + \frac{g}{\ell}$: a stable focus for light friction and a stable node for heavy friction. The upright equilibrium has $\lambda^2 + \frac{b}{m\ell^2}\lambda - \frac{g}{\ell}$, whose roots are real and of opposite sign: a saddle. One nonlinear portrait contains both pictures, a spiral sink at $\theta = 0$ and a saddle at $\theta = \pi$, stitched together far from either.

Part III — Transfer Functions

Lecture 4 · 14 September 2026

5 The Laplace Transform and Transfer Functions

5.1 Laplace transform: the part we need

The (one-sided) Laplace transform of a signal $f(t)$ is

$$F(s) = \mathcal{L}\{f\}(s) = \int_{0^-}^{\infty} f(t)\, e^{-st}\, \mathrm{d}t, \qquad s \in \mathbb{C}. \tag{16}$$

The lower limit is written 0^-: the integral starts *before* $t = 0$, so that an impulse sitting at the origin, and the initial state, are counted.

$\delta(t) \;\mapsto\; 1$	$u_s(t) \;\mapsto\; \dfrac{1}{s}$	$t\,u_s(t) \;\mapsto\; \dfrac{1}{s^2}$
$t^n u_s(t) \;\mapsto\; \dfrac{n!}{s^{\,n+1}}$	$e^{-\alpha t} \;\mapsto\; \dfrac{1}{s+\alpha}$	$f(t-a)\,u_s(t-a) \;\mapsto\; e^{-as}F(s)$
$\sin \omega t \;\mapsto\; \dfrac{\omega}{s^2+\omega^2}$	$\cos \omega t \;\mapsto\; \dfrac{s}{s^2+\omega^2}$	$f * g \;\mapsto\; F(s)\,G(s)$

Here u_s is the unit step. The pairs are only half of the toolkit; the other half is the list of *operations* the transform turns into algebra.

	time domain	s-domain
linearity	$af(t)+bg(t)$	$aF(s)+bG(s)$
differentiation	$\dot f(t)$	$sF(s)-f(0^-)$
integration	$\int_0^t f(\tau)\,\mathrm{d}\tau$	$F(s)/s$
time shift	$f(t-a)\,u_s(t-a)$, $a>0$	$e^{-as}F(s)$
frequency shift	$e^{-\alpha t}f(t)$	$F(s+\alpha)$
multiplication by t	$t\,f(t)$	$-F'(s)$
convolution	$(f*g)(t)=\int_0^t f(\tau)g(t-\tau)\,\mathrm{d}\tau$	$F(s)\,G(s)$

Of these, the one that does the real work is differentiation:

$$\mathcal{L}\{\dot f\}=sF(s)-f(0),\qquad \mathcal{L}\{\ddot f\}=s^2F(s)-sf(0)-\dot f(0). \tag{17}$$

The one line to remember is its consequence: *if all initial conditions vanish, differentiation becomes multiplication by s* — and, by the integration rule, integration becomes division by s. That single fact turns a linear differential equation into an algebraic one, and it is the entire reason the transfer function exists.

Remark 5.1 (reading the table). Two of the rows are worth seeing as pictures. The frequency shift says that damping a signal by $e^{-\alpha t}$ slides its transform to the left by α in the s-plane: $\sin\omega t$ has poles at $\pm j\omega$, and $e^{-\alpha t}\sin\omega t$ has poles at $-\alpha\pm j\omega$. The time shift says that a pure delay of a seconds multiplies by e^{-as} — a factor that is not a ratio of polynomials, which is why delays sit awkwardly with everything built on rational transfer functions.

5.2 Initial and final values

Two limits let one read the beginning and the end of a signal straight off its transform, without inverting it.

Proposition 5.2 (initial- and final-value theorems)**.** *Let f have Laplace transform F.*

(i) If F is strictly proper, then $f(0^+)=\lim_{s\to\infty}sF(s)$.

(ii) If every pole of $sF(s)$ has strictly negative real part, then $f(t)$ has a limit as $t\to\infty$, and

$$\lim_{t\to\infty}f(t)=\lim_{s\to 0}sF(s).$$

Both follow from the differentiation rule: $\int_0^\infty \dot f(t)e^{-st}\,\mathrm{d}t=sF(s)-f(0)$, and letting $s\to 0$ (the integral becomes $f(\infty)-f(0)$) or $s\to\infty$ (the integral vanishes) gives the two statements.

> **Warning (the hypothesis of the final-value theorem).** The condition on the poles is not decoration. For $F(s)=1/(s-1)$ the formula gives $\lim_{s\to0}sF(s)=0$, yet $f(t)=e^t$ grows without bound. Likewise $F(s)=\omega/(s^2+\omega^2)$ gives 0, but $\sin\omega t$ has no limit at all. The theorem only reports a final value that exists; it cannot decide whether one does. That question is answered by the poles, and it is the subject of stability.

Example 5.3 (the steady offset of the held mass)**.** The feedback law of §1.3 produced (4), $m\ddot{y} + c\dot{y} + k(y-h) = (\hat{m}-m)g$. Put $e = y - h$; with $e(0) = \dot{e}(0) = 0$ and a constant right-hand side $d = (\hat{m}-m)g$,

$$E(s) = \frac{d}{s\,(ms^2 + cs + k)}, \qquad \lim_{t\to\infty} e(t) = \lim_{s\to 0} sE(s) = \frac{d}{k}.$$

The poles of $sE(s)$ are the roots of $ms^2 + cs + k$, which lie in the left half-plane for $c, k > 0$, so the theorem applies, and it reproduces (5) without solving the differential equation.

5.3 The transfer function

Definition 5.4 (transfer function)**.** For a linear time-invariant system with input $u(t)$ and output $y(t)$ that is *initially at rest* (all initial conditions zero), the *transfer function* is

$$P(s) \;=\; \frac{\mathcal{L}\{y(t)\}}{\mathcal{L}\{u(t)\}} \;=\; \frac{Y(s)}{U(s)}.$$

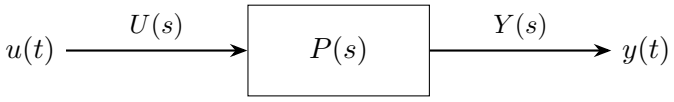


Figure 8. The transfer function as a block: $Y(s) = P(s)U(s)$, valid when the system starts at rest.

Equivalently, $P(s)$ is the Laplace transform of the impulse response: putting $u = \delta$ gives $U \equiv 1$ and hence $Y = P$. When several signals are around it is also written $T_{uy}(s)$, marking which signal is the input and which the output; for a single-input single-output system the plain $P(s)$ is unambiguous.

Remark 5.5 (why "initially at rest" is not a detail). Without that hypothesis the transformed equation carries extra terms $-sy(0) - \dot{y}(0) - \cdots$ coming from the initial conditions, and the ratio $Y(s)/U(s)$ then depends on how the system happened to start, not only on the system. It would not be a property of the plant, and calling it a transfer *function of the system* would be meaningless.

Example 5.6. Take $\ddot{y}+3\dot{y}+y = 2\dot{u}+u$. Transforming with zero initial conditions, $s^2Y+3sY+Y = 2sU + U$, so $P(s) = (2s+1)/(s^2+3s+1)$.

The recipe is visible in the example: in $a_n y^{(n)} + \cdots + a_1\dot{y} + a_0 y = b_m u^{(m)} + \cdots + b_1\dot{u} + b_0 u$, replace every $\mathrm{d}/\mathrm{d}t$ by s and solve for Y/U:

$$P(s) = \frac{b_m s^m + \cdots + b_1 s + b_0}{a_n s^n + \cdots + a_1 s + a_0}.$$

The degree of the denominator is the order of the system.

5.4 Example: the two-cart system, idealized

Strip the two-cart system of friction and of the coupling damper ($c = b = 0$):

$$m_1\ddot{p}_1 = k\,(p_2 - p_1) + u, \qquad m_2\ddot{p}_2 = k\,(p_1 - p_2), \qquad y = p_2. \tag{18}$$

Transforming with zero initial conditions, and writing $Y = P_2$,

$$m_1 s^2 P_1 = kY - kP_1 + U, \qquad m_2 s^2 Y = kP_1 - kY.$$

The second equation gives $P_1 = \dfrac{m_2 s^2 + k}{k}\, Y$. Substituting it into the first, $(m_1 s^2 + k)P_1 = kY + U$, yields

$$\frac{(m_1 s^2 + k)(m_2 s^2 + k)}{k}\, Y = kY + U \quad \Longrightarrow \quad \Big[(m_1 s^2 + k)(m_2 s^2 + k) - k^2\Big] Y = k\, U.$$

Since $(m_1 s^2 + k)(m_2 s^2 + k) - k^2 = s^2\big(m_1 m_2 s^2 + k(m_1 + m_2)\big)$,

$$\boxed{\frac{Y(s)}{U(s)} \;=\; \frac{k}{s^2\big(m_1 m_2\, s^2 + k(m_1 + m_2)\big)}} \tag{19}$$

Remark 5.7 (reading (19)). The factor s^2 in the denominator is the free rigid-body mode: with no friction, nothing ties the pair of carts to the ground, so a net impulse leaves them drifting forever. The remaining factor vanishes at $s = \pm j\sqrt{k(m_1 + m_2)/(m_1 m_2)}$, the frequency at which the two carts oscillate against each other. Both are on the imaginary axis — with $c = b = 0$ nothing damps, which is exactly why the model of Section 2 kept c and b.

> **Warning (two different two-cart models).** Equation (18) is *not* the model (6): the friction c and the damper b have been dropped to keep the algebra readable. Do not mix results derived from the two. When $c > 0$ the factor s^2 in (19) becomes $s(\,\cdot\,)$ — one pole moves off the origin and only one free integrator survives.

Remark 5.8 (the same computation with c and b kept). Carrying the full model (6) through the same steps gives, in matrix form,

$$\begin{bmatrix} m_1 s^2 + (b+c)s + k & -(bs+k) \\ -(bs+k) & m_2 s^2 + (b+c)s + k \end{bmatrix} \begin{bmatrix} P_1 \\ P_2 \end{bmatrix} = \begin{bmatrix} U \\ 0 \end{bmatrix},$$

so that, writing $\Delta(s) = \big[m_1 s^2 + (b+c)s + k\big]\big[m_2 s^2 + (b+c)s + k\big] - (bs+k)^2$, one gets $Y(s)/U(s) = (bs+k)/\Delta(s)$. Setting $b = c = 0$ recovers (19) exactly, which is a useful check on both. Two things are worth noticing. The coupling damper produces a finite zero at $s = -k/b$ — the numerator is $bs + k$, never 0. And with $c = 0$ but $b > 0$ one gets $\Delta(s) = s^2\big[m_1 m_2 s^2 + (m_1 + m_2)bs + (m_1 + m_2)k\big]$: the double pole at the origin survives, since it comes from the absence of *ground* friction, not from the absence of the damper.

5.5 From state space to transfer function

The two representations of Figure 3 are connected by a single computation. Start from (8) with $x(0) = 0$ and transform: $s\,X(s) = A\,X(s) + B\,U(s)$. Collect the X terms, $(sI - A)X(s) = B\,U(s)$, and invert: $X(s) = (sI - A)^{-1} B\,U(s)$. Feeding this into $Y = CX + DU$,

$$\boxed{Y(s) = \big[\,C\,(sI - A)^{-1}B + D\,\big]\,U(s), \qquad P(s) = C(sI - A)^{-1}B + D.} \tag{20}$$

For a SISO system this is a scalar rational function of s; in general it is a $p \times m$ matrix of them.

Remark 5.9 (keeping $x(0)$, and where the transfer function comes from). It is worth doing the computation once without assuming $x(0) = 0$, because it shows exactly what the zero-initial-condition hypothesis buys. Transforming $\dot{x} = Ax + Bu$ in full gives $sX(s) - x(0) = AX(s) + BU(s)$, hence $(sI - A)X(s) = x(0) + BU(s)$ and

$$Y(s) = \underbrace{C(sI - A)^{-1}x(0)}_{\text{free response}} + \underbrace{\big[C(sI - A)^{-1}B + D\big]U(s)}_{\text{forced response}}.$$

The transfer function is the second term alone. The first term does not disappear in reality; it is simply not part of the definition, which is why Definition 5.4 has to say "initially at rest".

Remark 5.10. Since $(sI-A)^{-1} = \operatorname{adj}(sI-A)/\det(sI-A)$, we may also write $P(s) = C\,\operatorname{adj}(sI-A)\,B/\det(sI-A) + D$, so every pole of $P(s)$ is an eigenvalue of A — though not necessarily the other way round, since factors can cancel. The state-space model can therefore carry information the transfer function does not, which is one reason this course does not work with transfer functions alone. Going the other way, from $P(s)$ back to a quadruple (A,B,C,D), is called *realization*; it is always possible for a proper P, but the answer is far from unique. That is why the arrow between the two boxes in Figure 3 points both ways.

Example 5.11 (a realization, step by step)**.** Realize $P(s) = \dfrac{s^2+s+2}{s^3+3s^2+2s-1}$. Split it through an intermediate signal $\bar{y}$:

$$U \;\longrightarrow\; \boxed{\frac{1}{s^3+3s^2+2s-1}} \;\longrightarrow\; \bar{Y} \;\longrightarrow\; \boxed{s^2+s+2} \;\longrightarrow\; Y.$$

The first block is the differential equation $\dddot{\bar{y}} + 3\ddot{\bar{y}} + 2\dot{\bar{y}} - \bar{y} = u$, which has no input derivatives, so the choice $x_1 = \bar{y}$, $x_2 = \dot{\bar{y}}$, $x_3 = \ddot{\bar{y}}$ of §3.1 turns it into

$$\dot{x}_1 = x_2, \qquad \dot{x}_2 = x_3, \qquad \dot{x}_3 = x_1 - 2x_2 - 3x_3 + u.$$

The second block is only differentiation of $\bar{y}$, and every derivative it needs is already a state: $y = \ddot{\bar{y}} + \dot{\bar{y}} + 2\bar{y} = 2x_1 + x_2 + x_3$. Hence

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & -2 & -3 \end{bmatrix}, \qquad B = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \qquad C = \begin{bmatrix} 2 & 1 & 1 \end{bmatrix}, \qquad D = 0.$$

The pattern is general: the denominator coefficients fill the last row of A (with signs reversed), and the numerator coefficients become C. As a check, $(sI-A)^{-1}B = (1,\ s,\ s^2)^{\mathsf{T}}/(s^3+3s^2+2s-1)$, and C times that is $P(s)$.

5.6 Poles, zeros, and properness

Every transfer function we shall meet is a ratio of polynomials,

$$G(s) = \frac{b_m s^m + \cdots + b_1 s + b_0}{s^n + a_{n-1}s^{n-1} + \cdots + a_1 s + a_0}, \tag{21}$$

a *rational* transfer function. Its *degree* is n, the degree of the denominator, and its *relative degree* is $n-m$. Comparing n with m gives four names:

$n \geq m$	proper
$n > m$	strictly proper
$n = m$	biproper
$n < m$	improper

Remark 5.12 (why improper is unwelcome). An improper transfer function has more derivatives on the output side than the input side, so realising it requires *differentiating* the input. A pure differentiator has $|j\omega| = \omega$: its gain grows without bound with frequency, so it amplifies exactly the high-frequency measurement noise one wants to suppress. Real plants are strictly proper, which is another way of saying that nothing responds instantaneously; controllers, which we build ourselves, are kept at least proper for the same reason.

Factoring numerator and denominator,

$$G(s) = \frac{N(s)}{D(s)} = k\,\frac{\prod_{i=1}^{m}(s - z_i)}{\prod_{i=1}^{n}(s - p_i)}, \tag{22}$$

where N and D are real polynomials. The $s \in \mathbb{C}$ with $D(s) = 0$ are the **poles** — there are n of them — and the s with $N(s) = 0$ are the **zeros**, of which there are m. Since N and D have *real* coefficients, any pole or zero that is not real arrives together with its complex conjugate. Marking them in the complex plane, poles with $\times$ and zeros with $\circ$, gives the *pole–zero map* (Figure 9).

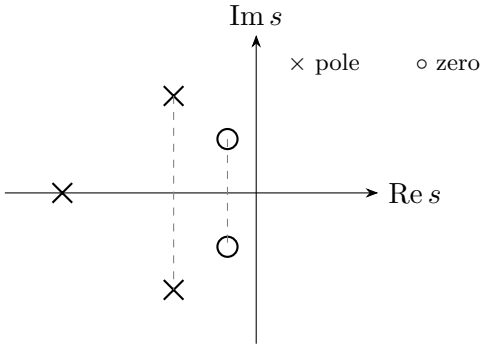


Figure 9. A pole–zero map. Because N and D have real coefficients, every non-real pole and zero is paired with its mirror image across the real axis (dashed); real ones stand alone.

5.7 Inverting by partial fractions

To go back from $Y(s)$ to $y(t)$ one almost never evaluates an inverse integral. Instead one splits a rational $Y(s)$ into a sum of simple terms, each of which is a line of the table of pairs. If Y is strictly proper with poles p_i, there are three kinds of term.

Distinct real poles. Each simple pole p_i contributes $A_i/(s - p_i)$, and its coefficient is found by the *cover-up rule*: multiply by $(s - p_i)$ and set $s = p_i$,

$$A_i = \big[(s - p_i)\,Y(s)\big]_{s=p_i}. \tag{23}$$

Each such term inverts to $A_i e^{p_i t}$.

A repeated pole. A pole of multiplicity r contributes r terms, $B_1/(s-p) + B_2/(s-p)^2 + \cdots + B_r/(s-p)^r$. The top coefficient B_r is found by cover-up with $(s-p)^r$; the lower ones by differentiating: $B_{r-k} = \frac{1}{k!}\frac{\mathrm{d}^k}{\mathrm{d}s^k}\big[(s-p)^r Y(s)\big]_{s=p}$. The term $B/(s-p)^j$ inverts to $B\,t^{j-1}e^{pt}/(j-1)!$ — a repeated pole brings a polynomial in t in front of the exponential.

A complex pair. Poles $-\alpha \pm j\omega$ are kept together as one real quadratic, $(s+\alpha)^2 + \omega^2$, over a linear numerator. Completing it into the two shifted pairs of the table,

$$\frac{C(s+\alpha) + D\omega}{(s+\alpha)^2 + \omega^2} \;\longmapsto\; e^{-\alpha t}\big(C\cos\omega t + D\sin\omega t\big),$$

gives a decaying (for $\alpha > 0$) oscillation. The real part of the poles sets the envelope, the imaginary part the frequency.

Example 5.13 (the three cases)**.** (a) $Y(s) = \dfrac{s+5}{(s+1)(s+2)} = \dfrac{A}{s+1} + \dfrac{B}{s+2}$. Cover-up gives $A = \frac{s+5}{s+2}\big|_{s=-1} = 4$ and $B = \frac{s+5}{s+1}\big|_{s=-2} = -3$, so $y(t) = 4e^{-t} - 3e^{-2t}$.

(b) $Y(s) = \dfrac{1}{s(s+1)^2} = \dfrac{A}{s} + \dfrac{B_1}{s+1} + \dfrac{B_2}{(s+1)^2}$. Cover-up gives $A = 1$ and $B_2 = \frac{1}{s}\big|_{s=-1} = -1$; differentiating, $B_1 = \frac{\mathrm{d}}{\mathrm{d}s}\frac{1}{s}\big|_{s=-1} = -1$. Hence $y(t) = 1 - e^{-t} - t\,e^{-t}$.

(c) $Y(s) = \dfrac{1}{s^2+2s+5} = \dfrac{1}{(s+1)^2+2^2} = \dfrac{1}{2}\cdot\dfrac{2}{(s+1)^2+2^2}$, so $y(t) = \frac{1}{2}e^{-t}\sin 2t$.

In (b) the two theorems of §5.2 check the answer: $\lim_{s\to\infty} sY(s) = 0 = y(0^+)$ and $\lim_{s\to 0} sY(s) = 1 = y(\infty)$.

5.8 Where the poles of the output come from

With the plant in hand, the response to an input is $y(t) = \mathcal{L}^{-1}\{G(s)\,U(s)\}$, and the way to read that inverse transform is to split it into partial fractions. Suppose it comes out as

$$Y(s) = \frac{A}{s-1} + \frac{B}{s-2} + \frac{C}{s+3} + \frac{Ds+E}{s^2+\omega^2}. \tag{24}$$

The first three denominators are poles of the *plant*; the last is a pole of the *input* — an input $\sin\omega t$ would contribute exactly that factor. Inverting term by term gives one time function per pole, so

the poles of the output are the poles of the plant together with the poles of the input.

The plant's own poles appear whatever we drive it with, which is why the question of whether they are well placed is a question about the plant alone.

5.9 Block diagrams and the feedback loop

Because $Y(s) = P(s)U(s)$ is plain multiplication, systems wired together combine by algebra. Three connections cover almost every diagram (Figure 10):

$$\underbrace{Q(s)P(s)}_{\text{series}}, \qquad \underbrace{P(s)+Q(s)}_{\text{parallel}}, \qquad \underbrace{\frac{G(s)}{1+G(s)H(s)}}_{\text{negative feedback}}. \tag{25}$$

The first two are immediate. The third is the one that matters, and it is worth deriving once. In the loop of Figure 10(c) the error is $E = R - HY$ and the output is $Y = GE$, so $Y = G(R - HY)$: the output appears on both sides, which is the algebraic trace of feedback. Collecting it, $(1+GH)Y = GR$, and the formula follows.

The trade-off against the equations of motion of §2 is worth stating. A block diagram makes *connecting* systems trivial — one multiplies, adds, or applies (25) — but it hides the physics: a block labelled $P(s)$ no longer says which masses, springs and dampers are inside it. The differential equation is the opposite: every term has a physical meaning, and wiring two systems together means rewriting the equations. The two are used side by side for that reason.

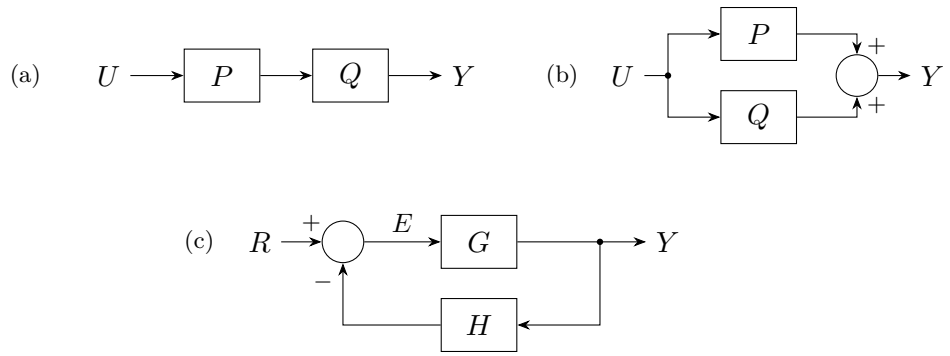


Figure 10. (a) Series: $Y = QPU$. (b) Parallel: $Y = (P+Q)U$. (c) Negative feedback: $Y = \frac{G}{1+GH}R$.

In the standard control loop the forward path is a controller followed by the plant, $G = P(s)C(s)$, and the output is fed back directly, $H = 1$. The reference r is what we want y to be, and the controller acts on the error $e = r - y$:

$$\frac{Y(s)}{R(s)} = \frac{P(s)C(s)}{1 + P(s)C(s)}, \qquad \frac{E(s)}{R(s)} = \frac{1}{1 + P(s)C(s)}. \tag{26}$$

Definition 5.14 (loop transfer function)**.** The product of all the blocks met in going once around the loop,

$$L(s) = P(s)\,C(s)$$

(more generally $G(s)H(s)$), is the *loop transfer function* — also called the *loop gain*, or the *open-loop* transfer function. The equation

$$1 + L(s) = 0 \tag{27}$$

is the *characteristic equation* of the closed loop.

Every closed-loop transfer function in (26) has the same denominator $1 + L$. Writing $L = N_L/D_L$ as a ratio of polynomials,

$$\frac{L}{1+L} = \frac{N_L}{D_L + N_L},$$

so the closed-loop poles are the roots of $D_L + N_L$, equivalently of (27). This is the precise sense in which *feedback moves poles*: the plant alone has the roots of D_L; closing the loop replaces them by the roots of $D_L + N_L$, and the controller C is our handle on the difference.

Example 5.15 (the held mass, as a loop)**.** Go back to §1.3 and measure everything from the target height, $e = h - y$. The plant from force to height is $P(s) = 1/(ms^2)$, and the feedback law (3) is, apart from the gravity-compensating constant, the controller $C(s) = k + cs$. Then

$$L(s) = \frac{k + cs}{ms^2}, \qquad 1 + L(s) = \frac{ms^2 + cs + k}{ms^2}.$$

The open loop has a double pole at the origin — a free mass, which drifts. The closed loop has the roots of $ms^2 + cs + k$, which for $c, k > 0$ lie in the left half-plane. The gains have become an *effective damping* c and an *effective stiffness* k that the free mass did not have, and that is the whole content of "feedback stabilised it".

Question 5.16. Given a plant P and a proposed controller C, is the closed loop stable? One answer is to compute the roots of $D_L + N_L$. Section 7 gives a second answer, which needs only the graph of $L(j\omega)$ — something one can measure — and which tells us not just *whether* the loop is stable but *how far* it is from losing stability.

Part IV — Frequency Response and Stability

6 Frequency Response and Bode Plots

6.1 Stable plants

Definition 6.1 (stable plant)**.** A plant is *stable* when every pole has strictly negative real part.

Remark 6.2 (why this section begins with stability)*.* Everything below evaluates P on the imaginary axis, at $s = j\omega$ — that is, it trades the Laplace transform for the Fourier transform — and that substitution is only legitimate when the Laplace integral actually converges there. For a stable plant it does, and $P(j\omega)$ is the Fourier transform of the impulse response — something one can measure by driving the plant with a sinusoid. If a pole sits in the right half-plane the imaginary axis lies outside the region of convergence, the response to a sinusoid grows without bound, and the symbol $P(j\omega)$ is not the frequency response of anything. Stability is a hypothesis of this whole section, not a side remark.

6.2 Sinusoid in, sinusoid out

Drive a stable plant with $\sin \omega t$ and wait for the transient to die away. What is left is a sinusoid *at the same frequency*:

$$\sin \omega t \;\longmapsto\; A \sin(\omega t + \theta), \qquad A = |P(j\omega)|, \qquad \theta = \angle P(j\omega). \tag{28}$$

Only two numbers change, and both are read off the single complex number $P(j\omega)$: its magnitude scales the amplitude, its argument shifts the phase. As functions of ω the pair is the *frequency response*. This is the practical reason transfer functions are worth having: it turns a differential equation into a gain and a phase shift, one frequency at a time.

Why (28) *holds.* The input transform is $\omega/(s^2+\omega^2)$, so by §5.8

$$Y(s) = P(s)\,\frac{\omega}{s^2+\omega^2} = \frac{as+b}{s^2+\omega^2} + \big(\text{terms from the poles of } P\big).$$

The plant is stable, so the second group dies out, and what remains is $y_{ss}(t) = a\cos\omega t + (b/\omega)\sin\omega t$. To find a and b, multiply both sides by $s - j\omega$ and set $s = j\omega$; the plant terms vanish and

$$P(j\omega)\,\frac{\omega}{2j\omega} = \frac{a\,j\omega + b}{2j\omega} \quad\Longrightarrow\quad P(j\omega) = \frac{b}{\omega} + ja.$$

So $b/\omega = |P(j\omega)|\cos\angle P(j\omega)$ and $a = |P(j\omega)|\sin\angle P(j\omega)$, and the sum-of-angles formula collapses y_{ss} to $|P(j\omega)|\sin\big(\omega t + \angle P(j\omega)\big)$. □

Example 6.3 (two frequencies at once)**.** Feed $u(t) = 2\sin t + 3\cos(2t+1)$ into $P(s) = 1/(s+1)$. By linearity each term is handled on its own, and a cosine obeys the same rule as a sine. At $\omega = 1$, $P(j) = 1/(1+j)$ has magnitude $1/\sqrt{2}$ and phase $-\pi/4$; at $\omega = 2$, $P(2j) = 1/(1+2j)$ has magnitude $1/\sqrt{5}$ and phase $-\arctan 2 \approx -63.4°$. Hence

$$y_{ss}(t) = \sqrt{2}\,\sin\Big(t - \frac{\pi}{4}\Big) + \frac{3}{\sqrt{5}}\cos\big(2t + 1 - \arctan 2\big).$$

The higher frequency is both attenuated more and delayed more: $1/(s+1)$ is a low-pass filter, which the Bode plot of the next subsection shows at a glance.

6.3 The Bode plot

The frequency response is drawn as two graphs against $\log_{10}\omega$: the magnitude in decibels,

$$20\log_{10}|P(j\omega)|,$$

and the phase $\angle P(j\omega)$. Together they are the *Bode plot*.

Both logarithms earn their place. A transfer function is a *product* of factors, and taking the logarithm of the magnitude turns that product into a *sum*, so each factor contributes its own piece and the whole is obtained by adding the pieces up. Putting $\log\omega$ on the horizontal axis then makes each of those pieces a straight line over most of its range.

6.4 Building the magnitude from the factors

Take, for example,

$$P(s) = \frac{K(s+z_1)(s+z_2)}{(s+p_1)(s+p_2)^2\big(s^2 + 2\zeta\omega_n s + \omega_n^2\big)}. \tag{29}$$

Taking $20\log_{10}$ of the magnitude turns every factor into a term:

$$\begin{aligned} 20\log|P| = 20\big(&\log|s+z_1| + \log|s+z_2| - \log|s+p_1| \\ &- 2\log|s+p_2| - \log|s^2 + 2\zeta\omega_n s + \omega_n^2|\big), \end{aligned} \tag{30}$$

evaluated at $s = j\omega$. Numerator factors enter with a plus, denominator factors with a minus, and a repeated factor enters with its multiplicity — hence the 2 in front of $\log|s+p_2|$.

Each first-order factor is then a single bend (Figure 11). For $s + z_1$ at $s = j\omega$,

$$20\log|j\omega + z_1| = 10\log_{10}\big(\omega^2 + z_1^2\big) \;\approx\; \begin{cases} 20\log z_1, & \omega \ll z_1, \\ 20\log\omega, & \omega \gg z_1, \end{cases}$$

so the asymptotes are a horizontal line and a line of slope +20 dB per decade, meeting at the *corner frequency* $\omega = z_1$. At the corner itself the true value is $20\log(z_1\sqrt{2}) = 20\log z_1 + 3.01$ dB: the curve passes 3 dB *above* the corner. A denominator factor gives the same picture upside down — slope −20 dB/decade and 3 dB *below* — and the repeated factor $(s+p_2)^2$ doubles both, to −40 dB/decade.

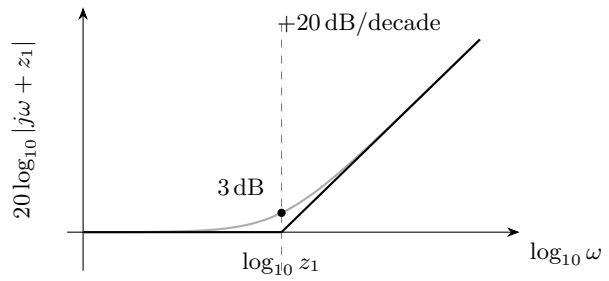


Figure 11. One first-order factor. The two straight asymptotes meet at the corner frequency $\omega = z_1$; the true magnitude (grey) runs 3 dB above the corner. A Bode magnitude plot is built by adding one such picture per factor.

6.5 Phase by adding angles

The phase is even easier, because the same factorisation is already a product of complex numbers. Writing each factor in polar form at $s = j\omega$,

$$P(j\omega) = \frac{\prod_i M_i e^{j\theta_i}}{\prod_j M'_j e^{j\theta'_j}}, \qquad \text{so} \qquad \angle P(j\omega) = \sum_i \theta_i - \sum_j \theta'_j.$$

Each factor's angle is measured separately and they are simply added and subtracted — the phase plot, like the magnitude plot, is assembled one factor at a time. For a first-order factor the angle is

$$\angle(j\omega + z_1) = \arctan\frac{\omega}{z_1},$$

which climbs from 0° at low frequency to 90° at high frequency and passes 45° exactly at the corner. Almost all of the change happens within a decade on either side, so the usual straight-line sketch holds 0° up to $z_1/10$, rises linearly in $\log\omega$ to 90° at $10z_1$, and stays there. A denominator factor contributes the negative of this.

Remark 6.4 (a geometric reading). $j\omega + z_1 = j\omega - (-z_1)$ is the vector from the zero at $-z_1$ to the point $j\omega$ on the imaginary axis. As ω runs up the axis, its length is the factor's magnitude and its angle is the factor's phase. Every factor of $P(j\omega)$ is such a vector, drawn from a pole or a zero to $j\omega$: the magnitude of P is the product of the zero-vector lengths divided by the product of the pole-vector lengths, and the phase is the sum of the zero-vector angles minus the sum of the pole-vector angles. The same picture, with $j\omega$ replaced by a point running around a closed curve, is the key to Section 7.

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6.6 The second-order system and the damping ratio

The quadratic factor in (29) deserves its own treatment: it is the one factor whose Bode plot the asymptotes do *not* describe well, and it is the transfer function of every damped oscillator. The standard second-order system is

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}, \qquad \omega_n > 0, \quad \zeta \geq 0, \tag{31}$$

normalised so that $G(0) = 1$. The two parameters have names: ω_n is the *natural frequency* and ζ the *damping ratio*. Both are forced on us by physics. A mass on a spring with a damper, $m\ddot{q} + b\dot{q} + kq = ku$, divided through by m is exactly (31) with

$$\omega_n = \sqrt{k/m}, \qquad \zeta = \frac{b}{2\sqrt{km}}, \tag{32}$$

so ω_n is the frequency at which the system would oscillate if there were no damping, and ζ measures the actual damping against the amount $2\sqrt{km}$ that just suppresses oscillation. Because the damping term is written $2\zeta\omega_n s$ and not $2\zeta s$, ζ is a pure number, and everything said below about ζ holds whatever the time scale of the system.

Poles. The roots of $s^2 + 2\zeta\omega_n s + \omega_n^2$ are

$$s = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1},$$

and the value of ζ sorts the system into four regimes:

$\zeta > 1$	overdamped	two distinct negative real poles
$\zeta = 1$	critically damped	a double pole at $-\omega_n$
$0 < \zeta < 1$	underdamped	$-\zeta\omega_n \pm j\omega_d$, $\quad \omega_d = \omega_n\sqrt{1-\zeta^2}$
$\zeta = 0$	undamped	$\pm j\omega_n$, on the imaginary axis

In the underdamped case, which is the interesting one, the poles lie on the circle $|s| = \omega_n$, and the line from the origin to either pole makes an angle θ with the negative real axis given by

$$\cos\theta = \zeta \tag{33}$$

(Figure 12). Distance from the origin is frequency; the angle is damping. Lines of constant ζ are rays from the origin.

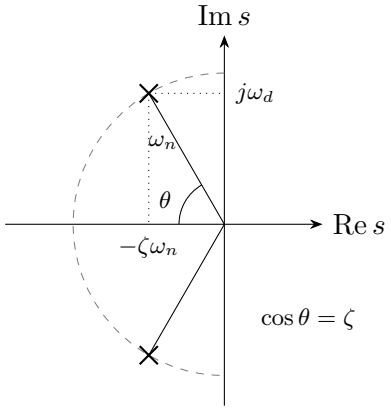


Figure 12. The underdamped pole pair of (31), drawn for $\zeta = 0.5$. The poles sit on the circle of radius ω_n; the angle θ they make with the negative real axis is fixed by the damping ratio alone.

Time response. For $0 < \zeta < 1$, the partial fractions of §5.7 applied to the unit step, $Y(s) = G(s)/s$, give

$$y(t) = 1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}}\,\sin\bigl(\omega_d t + \theta\bigr), \qquad \theta = \arccos\zeta. \tag{34}$$

Every piece of this is read off the pole: the real part $-\zeta\omega_n$ sets the decaying envelope, the imaginary part ω_d the frequency of the ringing (Figure 13). Small ζ gives a quick but badly ringing response; ζ near 1 a sluggish one that barely overshoots. Two numbers summarise the curve, the peak overshoot and the settling time:

$$M_p = e^{-\pi\zeta/\sqrt{1-\zeta^2}}, \qquad t_s \approx \frac{4}{\zeta\omega_n} \quad \text{(to within 2\%)}. \tag{35}$$

The overshoot depends on ζ alone, that is on the *angle* of the poles; the settling time on $\zeta\omega_n$ alone, that is on their *distance from the imaginary axis*.

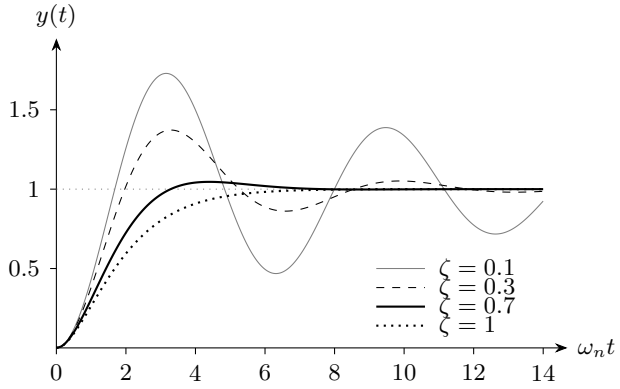


Figure 13. Unit-step responses of (31) against $\omega_n t$. Less damping means a higher first peak and slower-dying ringing; $\zeta = 1$ does not overshoot at all.

Remark 6.5 (the semester's goal, restated). Question 2.1 asks the carts to settle within 5 seconds. By (35), how fast a response settles and how far it overshoots are both statements about *where the dominant poles are*: settling within 5 s asks for $\zeta\omega_n \gtrsim 4/5$, and the overshoot is fixed by the angle θ. The open-loop two-cart system cannot be told where to put its poles. A closed loop can — that is what §5.9 showed feedback doing.

Frequency response. On the imaginary axis,

$$G(j\omega) = \frac{\omega_n^2}{(\omega_n^2 - \omega^2) + j\,2\zeta\omega_n\omega}. \tag{36}$$

With the normalised frequency $u = \omega/\omega_n$,

$$|G(j\omega)| = \frac{1}{\sqrt{(1-u^2)^2 + (2\zeta u)^2}}, \qquad \angle G(j\omega) = -\operatorname{atan2}\big(2\zeta u,\ 1-u^2\big).$$

The asymptotes are what one expects of a quadratic: for $u \ll 1$ the magnitude is 1 (0 dB) and the phase 0°; for $u \gg 1$ the magnitude is $1/u^2$, a line of slope -40 dB/decade, and the phase is $-180°$. They meet at $\omega = \omega_n$. What happens *at* the corner, however, is decided by ζ. Setting $u = 1$ in (36),

$$G(j\omega_n) = \frac{1}{j\,2\zeta}, \qquad |G(j\omega_n)| = \frac{1}{2\zeta}, \qquad \angle G(j\omega_n) = -90°.$$

The phase at the corner is $-90°$ for every ζ, but the magnitude is $1/(2\zeta)$, which is large when the damping is small: $\zeta = 0.1$ puts the curve 14 dB above the asymptotes (Figure 14). A first-order factor never strays more than 3 dB from its asymptotes; the quadratic can carry a tall *resonant*

peak. Setting the derivative of $|G|^2$ with respect to u to zero shows that for $\zeta < 1/\sqrt{2}$ the peak lies at

$$\omega_r = \omega_n\sqrt{1-2\zeta^2}, \qquad M_r = |G(j\omega_r)| = \frac{1}{2\zeta\sqrt{1-\zeta^2}}, \tag{37}$$

and that for $\zeta \geq 1/\sqrt{2} \approx 0.707$ there is no peak at all. The phase, meanwhile, falls from $0°$ to $-180°$ more and more abruptly as ζ shrinks; in the limit $\zeta \to 0$ it jumps by $180°$ at ω_n.

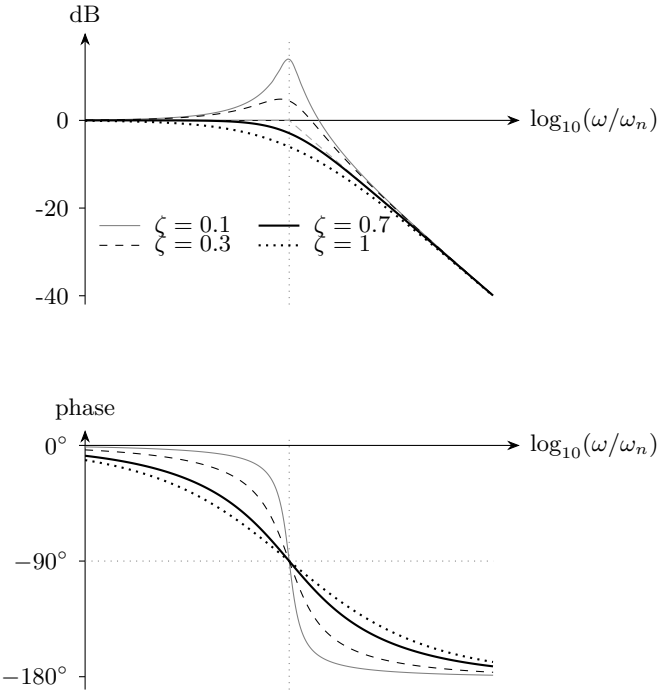


Figure 14. Bode magnitude (top) and phase (bottom) of the second-order system (31). The grey dashed lines in the top panel are the asymptotes. Every curve passes $-90°$ at $\omega = \omega_n$; the peak there, $20\log_{10}(1/2\zeta)$, grows without bound as $\zeta \to 0$.

Remark 6.6 (one fact, three pictures). Light damping shows up three times: as poles close to the imaginary axis, as ringing in the step response, and as a resonant peak in the Bode plot. These are the same fact seen three ways. A pole at $-\zeta\omega_n + j\omega_d$ lies a distance $\zeta\omega_n$ from the imaginary axis, so as $j\omega$ sweeps past it the pole-vector of Remark 6.4 becomes very short — and a short vector in the denominator is a large magnitude. Indeed at $\omega = \omega_d$ that vector has length $\zeta\omega_n$ and the other pole-vector has length about $2\omega_d$, which gives $|G| \approx \omega_n^2/(2\zeta\omega_n\omega_d) = M_r$.

Example 6.7 (resonance with no damping)**.** Push the limit $\zeta \to 0$ all the way. With $\zeta = 0$, $G(s) = \omega_n^2/(s^2+\omega_n^2)$ has its poles on the imaginary axis, so it is not stable and (28) does not apply. Drive it exactly at its natural frequency, $u = \sin\omega_n t$:

$$Y(s) = \frac{\omega_n^2}{s^2+\omega_n^2}\cdot\frac{\omega_n}{s^2+\omega_n^2} = \frac{\omega_n^3}{(s^2+\omega_n^2)^2}, \qquad y(t) = \tfrac{1}{2}\big(\sin\omega_n t - \omega_n t\cos\omega_n t\big).$$

The input's poles coincide with the plant's, the pole at $\pm j\omega_n$ becomes a double pole, and by §5.7 a double pole brings a factor of t: the oscillation grows linearly and without bound. This is resonance in its purest form, and the infinite peak of the $\zeta \to 0$ Bode curve is its frequency-domain shadow.

Remark 6.8 (dominant poles). Why study the second-order system so closely, when real plants have many poles? Because in a stable system each pole contributes a term $e^{p_i t}$, and the terms from poles far to the left die out long before those from poles near the imaginary axis. The poles closest to the axis — the *dominant* poles — therefore shape everything one sees after the first moments. When they are a complex pair well separated from the rest (and not nearly cancelled by a zero, which would make their residue small), an nth-order system behaves very nearly like the second-order system with those two poles. That is why ζ, ω_n, overshoot and settling time serve as the working vocabulary even for plants that are not second order.

6.7 A catalogue of elementary factors

Every rational transfer function is a product of the factors below, so every Bode plot is a sum of their pictures. The first-order factors are written in the normalised form that starts at 0 dB.

factor	magnitude	phase	at the corner
$K > 0$	flat at $20\log_{10} K$	$0°$	($K < 0$ adds $\pm 180°$)
s	$+20$ dB/dec, 0 dB at $\omega = 1$	$+90°$	—
$1/s$	-20 dB/dec, 0 dB at $\omega = 1$	$-90°$	—
$1 + s/z$	0, then $+20$ dB/dec from $\omega = z$	$0° \to +90°$	$+3$ dB, $+45°$
$1/(1 + s/p)$	0, then -20 dB/dec from $\omega = p$	$0° \to -90°$	-3 dB, $-45°$
$\dfrac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$	0, then -40 dB/dec from ω_n	$0° \to -180°$	$1/(2\zeta)$, $-90°$

Example 6.9 (a complete sketch)**.** Take $P(s) = \dfrac{10(s+1)}{s(s+10)} = \dfrac{1+s}{s\,(1+s/10)}$. It is a product of an integrator, a first-order zero with corner 1, and a first-order pole with corner 10. Reading left to right: below $\omega = 1$ only the integrator is active, so the magnitude falls at -20 dB/decade through 0 dB at $\omega = 1$ and the phase is $-90°$; at $\omega = 1$ the zero cancels the slope and the magnitude flattens at 0 dB while the phase rises toward $0°$; at $\omega = 10$ the pole brings the slope back to -20 dB/decade and the phase back toward $-90°$. The whole plot is three straight pieces joined at the corners 1 and 10.

7 The Nyquist Plot and Closed-Loop Stability

7.1 The question, restated

By §5.9, the closed loop with loop transfer function $L = PC$ is stable exactly when the characteristic equation $1 + L(s) = 0$ has no roots with $\operatorname{Re} s \geq 0$. Put

$$F(s) = 1 + L(s) = \frac{D_L(s) + N_L(s)}{D_L(s)}.$$

The *zeros* of F are the closed-loop poles — unknown, and exactly what we want to locate. The *poles* of F are the poles of L, the open-loop poles — known, since we know the plant and we chose the controller. The idea due to Nyquist is to count the unknown zeros in the right half-plane without computing them, by watching the phase of F as s travels around the boundary of that half-plane.

7.2 Counting by angles: the principle of the argument

Let Γ be a simple closed curve in the s-plane, traversed once *clockwise*, and let F be rational with no poles or zeros on Γ itself. As s goes around Γ, the point $F(s)$ traces a closed curve $F(\Gamma)$ in the F-plane (Figure 15). Factor F as in (22); by §6.5,

$$\angle F(s) = \angle k + \sum_i \angle(s - z_i) - \sum_j \angle(s - p_j).$$

Now follow a single factor. If z_i lies *inside* Γ, the vector $s - z_i$ from z_i to the moving point s turns once all the way round as s circles Γ — clockwise, so its angle changes by -2π. If z_i lies *outside*, the vector swings to and fro but comes back to where it started: net change 0. The same holds for each pole, with the opposite sign in $\angle F$. Adding up the factors:

Theorem 7.1 (principle of the argument)**.** *Let Z and P be the numbers of zeros and poles of F inside Γ, counted with multiplicity. As s traverses Γ once clockwise, $\angle F(s)$ changes by $-2\pi(Z - P)$ in total. That is, the curve $F(\Gamma)$ encircles the origin*

$$N = Z - P$$

times clockwise, a counterclockwise encirclement counting as -1.

For instance, put two points a, b inside Γ and a third point c outside. Then $F = s - a$ winds once clockwise around the origin; $F = (s - a)(s - b)$ winds twice, because θ_a and θ_b each turn through a full circle; and $F = (s - a)(s - b)(s - c)$ *still* winds twice, because θ_c only rocks within the wedge between the two tangents from c to Γ. A pole reverses the sense: $F = 1/(s - a)$ winds once *counterclockwise*, since $\angle F = -\theta_a$.

The angle count *is* the proof: it is the phase rule of §6.5 applied to a point that goes around a loop instead of up the imaginary axis. (In complex analysis the same count appears as $\frac{1}{2\pi j}\oint_\Gamma F'(s)/F(s)\,\mathrm{d}s = Z - P$ for a counterclockwise Γ: $F'/F = \sum_i 1/(s - z_i) - \sum_j 1/(s - p_j)$, and each term enclosed by Γ contributes a residue of $+1$ or -1.)

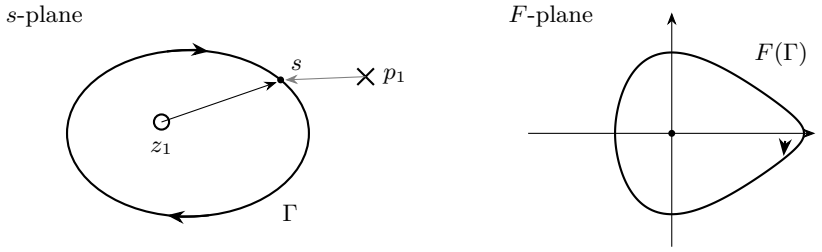


Figure 15. The principle of the argument. As s goes once clockwise around Γ, the vector $s - z_1$ from the enclosed zero turns once clockwise, while the vector $s - p_1$ from the pole outside only rocks back and forth. With one zero and no pole inside, the image $F(\Gamma)$ encircles the origin once clockwise: $N = Z - P = 1$.

7.3 The Nyquist contour and the Nyquist plot

To count right-half-plane roots, take Γ to be the boundary of the right half-plane. The *Nyquist contour* Γ_N runs up the imaginary axis from $-jR$ to $+jR$ and returns along the semicircle $|s| = R$ through the right half-plane, clockwise, with $R \to \infty$ (Figure 16). Its inside is the whole open right half-plane. If L has poles *on* the imaginary axis — an integrator $1/s$, say — Γ_N is not allowed to

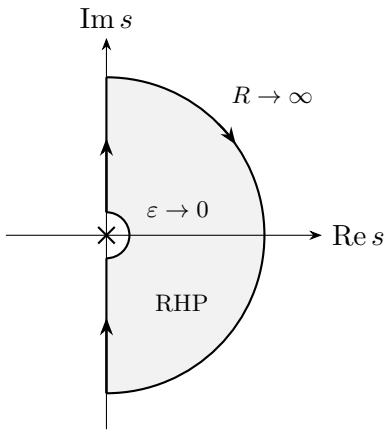


Figure 16. The Nyquist contour Γ_N, traversed clockwise: up the imaginary axis, then back along a semicircle of infinite radius. Its inside (shaded) is the open right half-plane. A pole on the imaginary axis — here at the origin — is avoided by a small detour into the right half-plane.

pass through them. It detours around each one on a small semicircle of radius $\varepsilon \to 0$ that bulges into the right half-plane, which leaves those poles outside.

The image of Γ_N under L is the *Nyquist plot* of L. Two facts make it easy to draw.

- If L is strictly proper, $L(s) \to 0$ as $|s| \to \infty$, so the whole large semicircle maps to the single point 0. The plot is then just the curve $L(j\omega)$, $-\infty < \omega < \infty$ — the frequency response, drawn in the complex plane instead of as two Bode graphs.
- Because L has real coefficients, $L(-j\omega) = \overline{L(j\omega)}$: the half for $\omega < 0$ is the mirror image, across the real axis, of the half for $\omega > 0$.

In practice one plots $L(j\omega)$ for $\omega \geq 0$ — read straight off the Bode plot, since the distance from the origin is $|L(j\omega)|$ and the angle is $\angle L(j\omega)$ — reflects it in the real axis, and adds whatever the small detours contribute.

7.4 The Nyquist stability criterion

Apply Theorem 7.1 to $F = 1 + L$ on Γ_N. The zeros of F inside Γ_N are the closed-loop poles in the right half-plane; the poles of F inside are the open-loop poles there. And $F(\Gamma_N)$ is the Nyquist plot of L shifted one unit to the right, so encircling the origin with F is the same as encircling the point -1 with L. That is the whole criterion.

Nyquist stability criterion. Let P be the number of poles of $L(s)$ in the open right half-plane, and let N be the number of times the Nyquist plot of L encircles the point -1 clockwise (counterclockwise encirclements counted negative). Then the number of closed-loop poles in the right half-plane is

$$Z = N + P.$$

The closed loop is stable if and only if $Z = 0$, that is $N = -P$, and the plot does not pass through -1 (if it does, the closed loop has a pole on the imaginary axis). In particular, if L is itself stable ($P = 0$), the closed loop is stable if and only if the Nyquist plot does not encircle -1.

Counted the other way round, the same statement reads: *the number of counterclockwise encirclements of* -1 *equals* $P - Z$.

Remark 7.2 (why this is worth the trouble). Two things set the criterion apart from simply computing the roots of $D_L + N_L$. First, it needs only $L(j\omega)$, and that can be *measured*: drive the open loop with sinusoids and record gain and phase, with no model of the plant at all. Second, it shows not only whether the loop is stable but how *close* it is to instability. A plot that passes near -1 can be pushed across it by a small change of gain or phase — by exactly the kind of modelling error that §1 said we must expect.

7.5 Examples

Example 7.3 (a stable open loop)**.** $L(s) = \dfrac{K}{(s+1)(s+2)(s+3)}$ with $K > 0$. All open-loop poles are in the left half-plane, so $P = 0$. At $s = j\omega$ the denominator is

$$(6 - 6\omega^2) + j\,(11\omega - \omega^3).$$

The plot starts at $L(0) = K/6$ on the positive real axis and moves clockwise through the fourth and third quadrants as the phase falls from 0° toward -270°. It crosses the negative real axis where the imaginary part of the denominator vanishes, at $\omega^2 = 11$, and there $L = K/(6 - 66) = -K/60$ (Figure 17, left, drawn for $K = 20$). If $K < 60$ the crossing lies to the right of -1 and there is no encirclement: $N = 0$, $Z = 0$, stable. If $K > 60$ the plot encircles -1 twice clockwise: $N = 2$, so two closed-loop poles lie in the right half-plane. At the boundary $K = 60$ the plot passes through -1 at $\omega = \sqrt{11}$, and indeed

$$D_L + N_L = s^3 + 6s^2 + 11s + 66 = (s + 6)(s^2 + 11)$$

has closed-loop poles exactly at $\pm j\sqrt{11}$ — the frequency at which the plot meets -1.

Example 7.4 (an integrator in the loop)**.** $L(s) = \dfrac{K}{s(s+1)}$ with $K > 0$. The pole at the origin lies on the imaginary axis, so Γ_N detours around it on $s = \varepsilon e^{j\phi}$ with ϕ running from -90° to $+90^\circ$. There $L(s) \approx K/s = (K/\varepsilon)e^{-j\phi}$: a semicircle of enormous radius K/ε, swept from $+90^\circ$ to -90°, that is clockwise through the right half of the plane. On the imaginary axis,

$$L(j\omega) = -\frac{K}{1+\omega^2} - j\,\frac{K}{\omega(1+\omega^2)},$$

so for $\omega > 0$ the plot rises from far below along the asymptote $\operatorname{Re} L = -K$, stays in the third quadrant, and enters the origin with phase -180° as $\omega \to \infty$. Its imaginary part never vanishes, so it never crosses the negative real axis and -1 is never encircled. The detoured pole is not counted, so $P = 0$; $N = 0$; hence $Z = 0$ and the loop is stable for every $K > 0$. Directly: $D_L + N_L = s^2 + s + K$, whose roots have negative real part for all $K > 0$.

The detour always behaves as in this example. Near a simple pole p on the imaginary axis, $L(s) \approx c/(s - p)$, and as the detour turns $s - p$ through $+180^\circ$ (from pointing down to pointing up, by way of the right), the image turns through -180° at infinite radius: half a clockwise circle. A double pole on the axis gives a full clockwise circle, -360° — enough, by itself, to wrap around -1.

Example 7.5 (a pair of poles on the imaginary axis)**.** $L(s) = \dfrac{1}{s^2 + 4} = \dfrac{1}{(s - 2j)(s + 2j)}$. Both poles lie on the axis and are detoured, so $P = 0$. Away from them, $L(j\omega) = 1/(4 - \omega^2)$ is real: it

runs along the positive real axis from $\frac{1}{4}$ out to $+\infty$ as $|\omega| \to 2$, and along the negative real axis from $-\infty$ back to 0 for $|\omega| > 2$. Each detour contributes a clockwise half-circle at infinity — the one at $+2j$ through the lower half-plane, the one at $-2j$ through the upper — and together they close the plot into an infinitely large circle. The whole negative real axis is covered, so the plot passes *through* -1, at $4 - \omega^2 = -1$, i.e. $\omega = \sqrt{5}$. By the criterion the closed loop has a pole on the imaginary axis, and indeed $1 + L = 0$ means $s^2 + 5 = 0$, $s = \pm j\sqrt{5}$: a sustained oscillation at $\sqrt{5}$ rad/s, neither decaying nor growing.

Example 7.6 (an unstable open loop)**.** $L(s) = \dfrac{K}{s-1}$ with $K > 0$. Now the open loop has a pole at $s = 1$, so $P = 1$. On the imaginary axis

$$L(j\omega) = \frac{K}{j\omega - 1} = -\frac{K}{1+\omega^2} - j\,\frac{K\omega}{1+\omega^2},$$

which traces the circle of radius $K/2$ centred at $-K/2$: from the origin ($\omega = -\infty$) through $-\frac{K}{2} + j\frac{K}{2}$, then $-K$ ($\omega = 0$), then $-\frac{K}{2} - j\frac{K}{2}$, and back to the origin — counterclockwise (Figure 17, right). If $K > 1$ the circle contains -1 and goes round it once counterclockwise: $N = -1$, so $Z = N + P = 0$ and the loop is stable. If $K < 1$ it misses -1: $N = 0$, $Z = 1$, unstable. Directly, $D_L + N_L = s - 1 + K$, stable exactly when $K > 1$. An unstable plant *needs* an encirclement to be stabilised; "the plot must not encircle -1" is the rule only for stable open loops.

Remark 7.7 (reading the criterion off a Bode plot). When L is stable ($P = 0$) and its phase falls through $-180°$ just once, at a frequency ω_{pc}, the Nyquist plot for $\omega > 0$ crosses the negative real axis just once, at the point $-|L(j\omega_{pc})|$. It encircles -1 exactly when that point lies to the left of -1. So the criterion becomes a statement one can check on the Bode plot:

$$\text{closed loop stable} \quad \Longleftrightarrow \quad |L(j\omega_{pc})| < 1 \quad \text{where } \angle L(j\omega_{pc}) = -180°.$$

By the time the phase has turned the loop signal fully upside down, the gain must already be below one. In Example 7.3 the phase reaches $-180°$ at $\omega_{pc} = \sqrt{11}$, where $|L| = K/60$ — the same condition $K < 60$ again. If the gain is too high there, it can be reduced (the idea behind a lag compensator), or the phase can be pushed back up so that $-180°$ is reached later (a lead compensator).

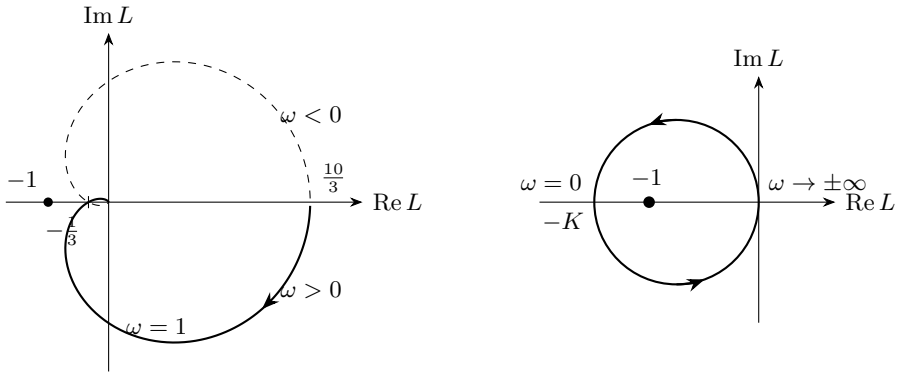


Figure 17. Left: the Nyquist plot of Example 7.3 for $K = 20$ (solid $\omega > 0$, dashed $\omega < 0$). It crosses the negative real axis at $-K/60 = -\frac{1}{3}$, to the right of -1, so -1 is not encircled and the closed loop is stable. Right: Example 7.6 for $K = 1.5$. The open loop is unstable ($P = 1$), and the plot goes once counterclockwise around -1 ($N = -1$), which is exactly what makes the closed loop stable.

Part V — Linear Algebra for State Space

Lectures 7–10 · 23 September – 7 October 2026

8 Vector Spaces, Rank, and Linear Equations

A state is a point in $\mathbb{R}^n$, an input pushes it along certain directions, and an output reports certain combinations of its coordinates. Everything the rest of the course does with $\dot{x} = Ax + Bu$ is therefore a statement about vectors, subspaces and matrices. This Part collects the linear algebra that will be used, in the order in which it is needed.

8.1 Matrices, read by columns

A matrix $A \in \mathbb{R}^{m\times n}$ has m rows and n columns; it is *square* if $m = n$, *tall* if $m > n$ and *fat* if $m < n$. Its transpose $A^{\mathsf{T}} \in \mathbb{R}^{n\times m}$ satisfies $(ABC)^{\mathsf{T}} = C^{\mathsf{T}}B^{\mathsf{T}}A^{\mathsf{T}}$, and for complex matrices the *Hermitian* (conjugate) transpose is $A^H = \overline{A}^{\mathsf{T}}$. The product $C = AB$ of $A \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{n\times k}$ has entries $c_{ij} = \sum_l a_{il}b_{lj}$, but three other ways of reading it are far more useful. Write $a_1, \dots, a_n$ for the columns of A and $b_1^{\mathsf{T}}, \dots, b_n^{\mathsf{T}}$ for the rows of B.

(i) $Ax = x_1a_1 + \cdots + x_na_n$: a matrix times a vector is a *linear combination of the columns*.

(ii) $y^{\mathsf{T}}B = y_1b_1^{\mathsf{T}} + \cdots + y_nb_n^{\mathsf{T}}$: a row vector times a matrix is a linear combination of the rows.

(iii) $AB = \sum_{i=1}^{n} a_i\, b_i^{\mathsf{T}}$: a product is a *sum of outer products*, each $a_ib_i^{\mathsf{T}}$ a matrix of rank one.

Reading (i) is the one to keep in mind throughout: asking whether $Ax = b$ can be solved is asking whether b is a combination of the columns of A.

8.2 Vector spaces, span, basis, dimension

Definition 8.1 (vector space)**.** A set S of vectors is a *vector space* (a *subspace* of $\mathbb{R}^n$) if it is closed under addition and scalar multiplication: $x, y \in S$ and $a, b \in \mathbb{R}$ imply $ax + by \in S$.

Taking $a = b = 0$ shows that every vector space contains the origin. A line or a plane *through the origin* is a subspace; one that misses the origin is not.

A *linear combination* of $a_1, \dots, a_k$ is $\sum_i c_i a_i$, and the vectors are *linearly independent* if $\sum_i c_i a_i = 0$ forces every $c_i = 0$ — no one of them can be built from the others; they all point in genuinely different directions. The set of all linear combinations is the *span*,

$$\operatorname{span}\{a_1, \dots, a_k\} = \Big\{ \sum_{i=1}^{k} c_i a_i \ : \ c_i \in \mathbb{R} \Big\},$$

and it is a vector space: adding two combinations, or scaling one, gives another combination. A *basis* of a vector space is a set of linearly independent vectors that spans it. A basis is not unique, but the number of vectors in every basis is the same; that number is the *dimension*. In the plane $x_3 = 0$ of $\mathbb{R}^3$, a single vector cannot span the plane, while any two non-parallel vectors in it form a basis, so the plane has dimension 2. A basis whose vectors are mutually perpendicular and of unit length is *orthonormal*.

There are two ways to describe a subspace, and both will be used.

- **By building it up** as a span. The plane $x_3 = 0$ is $\operatorname{span}\{e_1, e_2\}$; each independent generator adds one to the dimension ($0 \to 2$).
- **By cutting it down** with equations. The same plane is $\{x \in \mathbb{R}^3 : [0\ 0\ 1]\, x = 0\}$; each independent constraint removes one dimension ($3 \to 2$).

The line through the origin at 45° in the x_2x_3-plane is $\operatorname{span}\{(0, 1, 1)\}$, or equally $\{x \in \mathbb{R}^3 : [1\ 0\ 0]\, x = 0,\ [0\ 1\ -1]\, x = 0\}$: two independent constraints leave $3 - 2 = 1$ dimension.

8.3 Range, null space, and the equation $Ax = b$

Two subspaces belong to every matrix $A \in \mathbb{R}^{m \times n}$, one of each kind:

$$\mathcal{R}(A) = \{Ax : x \in \mathbb{R}^n\} \subseteq \mathbb{R}^m, \qquad \mathcal{N}(A) = \{x \in \mathbb{R}^n : Ax = 0\} \subseteq \mathbb{R}^n. \tag{38}$$

By reading (i), the *range* $\mathcal{R}(A)$ is the span of the columns — it is also called the *column space*, a description by building up. The *null space* $\mathcal{N}(A)$ is a description by cutting down. The *rank* of A is the number of linearly independent columns, which is always equal to the number of linearly independent rows, and it is the dimension of $\mathcal{R}(A)$.

The two subspaces answer the two questions one asks about a linear equation $Ax = b$:

- **Existence.** A solution exists if and only if $b \in \mathcal{R}(A)$ — if and only if b is a combination of the columns.
- **Uniqueness.** If $Ax = b$ and $Ax^* = b$, then $A(x - x^*) = 0$, so the two solutions differ by an element of $\mathcal{N}(A)$. A solution is unique if and only if $\mathcal{N}(A) = \{0\}$, which by reading (i) says that $x_1 a_1 + \cdots + x_n a_n = 0$ only for $x = 0$: the columns are linearly independent.

A matrix whose columns are independent, $\operatorname{rank} A = n$, is said to have *full column rank*, and then $\mathcal{N}(A) = \{0\}$. Existence and uniqueness are different questions with different answers, and it pays to keep them apart.

Remark 8.2 (counting dimensions). The cutting-down count above is a general fact. For $A \in \mathbb{R}^{m \times n}$,

$$\dim \mathcal{R}(A) + \dim \mathcal{N}(A) = n,$$

the *rank–nullity* relation: of the n input directions, $\operatorname{rank} A$ of them reach independent outputs and the remaining $n - \operatorname{rank} A$ are sent to zero.

Example 8.3 (the equilibria of the two carts)**.** For the two-cart matrix A of (10), the states that stay put when no force is applied are the solutions of $Ax = 0$. Rows 1 and 3 give $x_2 = x_4 = 0$: both carts at rest. Row 2 then reads $\frac{k}{m_1}(x_3 - x_1) = 0$, so $x_1 = x_3$, and row 4 says the same. Hence

$$\mathcal{N}(A) = \operatorname{span}\{(1,0,1,0)^{\mathsf{T}}\}, \qquad \operatorname{rank} A = 3.$$

Any position with both carts equally far along and the spring unstretched is an equilibrium — there is no ground spring to pull them back. In particular the target of Question 2.1, both carts 1 m ahead and at rest, $x = (1,0,1,0)^{\mathsf{T}}$, is an equilibrium of the uncontrolled system: once there, the carts stay there with $u = 0$. Keeping the damper b changes nothing, since every b-term is multiplied by a velocity.

8.4 Inverse and determinant

A square matrix $A \in \mathbb{R}^{n\times n}$ is *invertible* if there is A^{-1} with $AA^{-1} = A^{-1}A = I$. Writing $A^{-1} = [x_1 \ \cdots \ x_n]$, the equation $AA^{-1} = I$ is the n equations $Ax_i = e_i$, so an inverse exists exactly when every e_i, and hence all of $\mathbb{R}^n$, lies in $\mathcal{R}(A)$; for a square matrix that happens exactly when $\mathcal{N}(A) = \{0\}$. The following are therefore equivalent: A is invertible; $\mathcal{R}(A) = \mathbb{R}^n$; $\mathcal{N}(A) = \{0\}$; the columns are independent; $\operatorname{rank} A = n$; $\det A \neq 0$; 0 is not an eigenvalue of A.

For a 2×2 matrix,

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix} = \frac{\operatorname{adj} A}{\det A},$$

and the formula $A^{-1} = \operatorname{adj} A / \det A$ holds in every dimension; it is the formula behind $(sI - A)^{-1} = \operatorname{adj}(sI - A)/\det(sI - A)$ of §5.5. The determinant satisfies $\det A^{\mathsf{T}} = \det A$ and $\det(ABC) = \det A \det B \det C$.

Proposition 8.4 (Schur complement)**.** *Let* $M = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$ *with* A *and* D *square. If* A *is invertible,*

$$\det M = \det A \, \det\bigl(D - CA^{-1}B\bigr);$$

if D *is invertible,* $\det M = \det D \, \det\bigl(A - BD^{-1}C\bigr)$.

Proof. Block elimination: $\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} I & 0 \\ CA^{-1} & I \end{bmatrix}\begin{bmatrix} A & B \\ 0 & D - CA^{-1}B \end{bmatrix}$. The first factor has determinant 1 and the second is block triangular. The other case is symmetric. □

The matrix $D - CA^{-1}B$ is the *Schur complement* of A in M — what is left of D after the block A has been eliminated, exactly as in Gaussian elimination on scalars.

9 Coordinates, Eigenvalues, and Diagonalization

9.1 Changing state coordinates

The state vector of a system is a choice. Positions and velocities of each cart are one choice; the position of the pair and the stretch of the spring would be another. Any invertible $T \in \mathbb{R}^{n\times n}$ defines new coordinates

$$z(t) = T\,x(t).$$

Substituting $x = T^{-1}z$ into (8),

$$\dot{z} = TAT^{-1}\,z + TB\,u, \qquad y = CT^{-1}\,z + D\,u. \tag{39}$$

The quadruple (A, B, C, D) becomes $(TAT^{-1}, TB, CT^{-1}, D)$, a *similarity transformation*. The physics has not changed, so neither should anything one can measure — and indeed the transfer function is unchanged:

$$CT^{-1}\big(sI - TAT^{-1}\big)^{-1}TB + D = CT^{-1}\big(T(sI - A)T^{-1}\big)^{-1}TB + D = C(sI - A)^{-1}B + D,$$

using $sI = T(sI)T^{-1}$. This is one reason the realization of §5.5 is far from unique: every invertible T turns one realization into another with the same transfer function.

9.2 Eigenvalues and eigenvectors

Definition 9.1 (eigenvalue, eigenvector)**.** A number λ is an *eigenvalue* of $A \in \mathbb{R}^{n\times n}$, with *eigenvector* $v \neq 0$, if $Av = \lambda v$.

An eigenvector is a direction that A does not turn, only stretches by the factor λ. Rewriting $Av = \lambda v$ as $(\lambda I - A)v = 0$ shows that λ is an eigenvalue exactly when $\mathcal{N}(\lambda I - A) \neq \{0\}$, i.e. when $\lambda I - A$ is not invertible:

$$\det(\lambda I - A) = \lambda^n + a_{n-1}\lambda^{n-1} + \cdots + a_1\lambda + a_0 = 0. \tag{40}$$

The left side is the *characteristic polynomial*, of degree n, so A has n eigenvalues counted with multiplicity — complex ones in conjugate pairs, since the coefficients are real. Every nonzero vector in $\mathcal{N}(\lambda I - A)$ is an eigenvector, and that subspace is the *eigenspace* of λ.

Eigenvalues do not depend on the coordinates: by (39),

$$\det\big(\lambda I - TAT^{-1}\big) = \det\big(T(\lambda I - A)T^{-1}\big) = \det T\,\det(\lambda I - A)\,\det T^{-1} = \det(\lambda I - A),$$

so the whole characteristic polynomial is invariant. And the denominator of $P(s) = C\,\mathrm{adj}(sI - A)B/\det(sI - A) + D$ is exactly $\det(sI - A)$: *the eigenvalues of A are the poles of the transfer function*, except where a factor cancels against the numerator. The two descriptions of Figure 3 meet here.

9.3 Diagonalization

Suppose A has n linearly independent eigenvectors $v_1, \ldots, v_n$ with eigenvalues $\lambda_1, \ldots, \lambda_n$. Stacking the n equations $Av_i = \lambda_i v_i$ side by side,

$$A\,[v_1\ \cdots\ v_n] = [\lambda_1 v_1\ \cdots\ \lambda_n v_n] = [v_1\ \cdots\ v_n]\begin{bmatrix}\lambda_1 & & \\ & \ddots & \\ & & \lambda_n\end{bmatrix},$$

that is

$$AV = V\Lambda, \qquad V^{-1}AV = \Lambda, \qquad A = V\Lambda V^{-1}. \tag{41}$$

V is invertible precisely because its columns are independent. In the language of §9.1, the coordinates $z = V^{-1}x$ make the state matrix diagonal.

When are there n independent eigenvectors? Two counts decide it. The *algebraic multiplicity* of an eigenvalue is how many times it is a root of (40); its *geometric multiplicity* is $\dim\mathcal{N}(\lambda I - A)$, the number of independent eigenvectors it has. Always $1 \leq$ geometric $\leq$ algebraic.

- Eigenvectors belonging to *distinct* eigenvalues are linearly independent. So if all n eigenvalues are distinct, A is diagonalizable.

- In general, A is diagonalizable if and only if the geometric multiplicity equals the algebraic multiplicity for every eigenvalue.

Example 9.2 (equal multiplicities, and not)**.** Both

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 1 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix}$$

have eigenvalues $1, 2, 2, 3$; the eigenvalue 2 has algebraic multiplicity 2 in each. For A, $\mathcal{N}(2I - A) = \operatorname{span}\{e_2, e_3\}$ has dimension 2: two independent eigenvectors, and A is (already) diagonal. For B,

$$2I - B = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}, \qquad \mathcal{N}(2I - B) = \operatorname{span}\{e_2\},$$

of dimension 1: the two copies of the eigenvalue share a single eigenvector direction, V cannot be made invertible, and B is not diagonalizable. The single off-diagonal 1 is what makes the difference.

Example 9.3 (the two carts, decoupled)**.** Take equal masses $m_1 = m_2 = m$ (and $b = 0$, as in (10)). Adding and subtracting the two equations of motion (6),

$$m(\ddot{p}_1 + \ddot{p}_2) = u - c(\dot{p}_1 + \dot{p}_2), \qquad m(\ddot{p}_1 - \ddot{p}_2) = u - 2k(p_1 - p_2) - c(\dot{p}_1 - \dot{p}_2).$$

So the coordinates $z = (p_1 + p_2,\ \dot{p}_1 + \dot{p}_2,\ p_1 - p_2,\ \dot{p}_1 - \dot{p}_2)$, i.e. $z = Tx$ with

$$T = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \end{bmatrix}, \qquad TAT^{-1} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & -\frac{c}{m} & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -\frac{2k}{m} & -\frac{c}{m} \end{bmatrix}, \qquad TB = \frac{1}{m} \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \end{bmatrix},$$

split the system into two independent pieces: the motion of the pair as a whole, which feels only friction, and the oscillation of the carts against each other, which feels friction and twice the spring. The eigenvalues can now be read off block by block: 0 and $-c/m$ from the first, and the roots of $\lambda^2 + \frac{c}{m}\lambda + \frac{2k}{m}$ from the second. For the parameters used in the first problem set ($m = 1$, $k = 4$, $c = 0.4$) they are 0, -0.4 and $-0.2 \pm 2.8213j$. The eigenvalue 0 belongs to the eigenvector $(1, 0, 1, 0)^{\mathsf{T}}$ of Example 8.3: the free drift of the pair, which nothing in the open loop holds back.

9.4 Symmetric matrices

Symmetric matrices, $A = A^{\mathsf{T}}$, are the best behaved of all, and they will return in §11.

Theorem 9.4 (spectral theorem)**.** *Let $A = A^{\mathsf{T}} \in \mathbb{R}^{n \times n}$. Then every eigenvalue of A is real, eigenvectors belonging to distinct eigenvalues are orthogonal, and one can choose n orthonormal eigenvectors. With them as the columns of V,*

$$V^{\mathsf{T}}V = I, \qquad V^{-1} = V^{\mathsf{T}}, \qquad A = V\Lambda V^{\mathsf{T}} = \sum_{i=1}^{n} \lambda_i\, v_i v_i^{\mathsf{T}}.$$

Orthogonality. If $Av_i = \lambda_i v_i$ and $Av_j = \lambda_j v_j$, then $\lambda_i\, v_j^{\mathsf{T}} v_i = v_j^{\mathsf{T}} A v_i = v_j^{\mathsf{T}} A^{\mathsf{T}} v_i = (Av_j)^{\mathsf{T}} v_i = \lambda_j\, v_j^{\mathsf{T}} v_i$, so $(\lambda_i - \lambda_j)\, v_j^{\mathsf{T}} v_i = 0$, and $v_j^{\mathsf{T}} v_i = 0$ when $\lambda_i \neq \lambda_j$. □

Within an eigenspace of dimension k one can always pick k orthonormal vectors, and eigenspaces of different eigenvalues are orthogonal to one another, so all the choices together give an orthonormal set. For a symmetric matrix the inverse of V costs nothing — it is the transpose — and $V^{\mathsf{T}}AV = \Lambda$ is much easier to compute than $V^{-1}AV$. The last form, $A = \sum \lambda_i v_i v_i^{\mathsf{T}}$, writes A as a sum of rank-one pieces, one along each eigen-direction.

9.5 The Jordan form

A matrix with too few eigenvectors, such as B above, cannot be diagonalized, but every square matrix can be brought to the next best thing.

Theorem 9.5 (Jordan form)**.** *Every $A \in \mathbb{R}^{n\times n}$ satisfies $AV = VJ$ for some invertible (possibly complex) V, where J is block diagonal with* Jordan blocks

$$J_k(\lambda) = \begin{bmatrix} \lambda & 1 & & \\ & \lambda & \ddots & \\ & & \ddots & 1 \\ & & & \lambda \end{bmatrix}$$

on the diagonal. A is diagonalizable exactly when every block is 1×1.

To see what the columns of V are, take a 3×3 example,

$$AV = VJ, \qquad J = \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 1 \\ 0 & 0 & \lambda_2 \end{bmatrix}, \qquad [Av_1\ Av_2\ Av_3] = [\lambda_1 v_1\ \ \lambda_2 v_2\ \ \lambda_2 v_3 + v_2].$$

The first two columns are ordinary eigenvectors. The third satisfies

$$(A - \lambda_2 I)\, v_3 = v_2,$$

so it is not an eigenvector — $A - \lambda_2 I$ does not annihilate it, but sends it onto the eigenvector v_2. Such a v_3 is a *generalized eigenvector*. A generalized eigenvector exists exactly when some eigenvalue has algebraic multiplicity greater than its geometric multiplicity; it supplies the direction that the missing eigenvector would have provided.

10 The Matrix Exponential and the State Response

10.1 Solving $\dot{x} = Ax$

For a scalar, $\dot{x} = ax$ has the solution $x(t) = e^{at}x_0$. The same is true for a system, once the exponential of a matrix is defined by the same series.

Definition 10.1 (matrix exponential)**.** For $A \in \mathbb{R}^{n\times n}$,

$$e^{At} = I + At + \frac{(At)^2}{2!} + \frac{(At)^3}{3!} + \cdots .$$

The series converges for every A and t. Differentiating term by term,

$$\frac{\mathrm{d}}{\mathrm{d}t} e^{At} = A + A^2 t + \frac{A^3 t^2}{2!} + \cdots = A\Big(I + At + \frac{(At)^2}{2!} + \cdots\Big) = Ae^{At},$$

so $x(t) = e^{At}x_0$ satisfies $\dot{x} = Ax$ and $x(0) = x_0$: it is *the* solution. The properties follow the scalar ones, with one exception:

$$e^{A\cdot 0} = I, \qquad e^{At_1}e^{At_2} = e^{A(t_1+t_2)}, \qquad \left(e^{At}\right)^{-1} = e^{-At}, \qquad e^{At}e^{Bt} \neq e^{(A+B)t} \text{ unless } AB = BA. \tag{42}$$

The third follows from the second with $t_2 = -t_1$; in particular e^{At} is always invertible. The last fails because the series multiply out with AB and BA appearing in different orders. In MATLAB the matrix exponential is `expm`, not `exp` (which exponentiates entry by entry): the position of the second cart at $t = 10$ from x_0 is `[0 0 1 0]*expm(A*10)*x0`.

10.2 The state response

With an input, multiply $\dot{x} - Ax = Bu$ by e^{-At}. The left side becomes an exact derivative, $e^{-At}\dot{x} - e^{-At}Ax = \frac{\mathrm{d}}{\mathrm{d}t}\big(e^{-At}x(t)\big)$, and integrating from 0 to t gives $e^{-At}x(t) - x_0 = \int_0^t e^{-A\tau}Bu(\tau)\,\mathrm{d}\tau$. Multiplying by e^{At}:

$$\boxed{x(t) = e^{At}x_0 + \int_0^t e^{A(t-\tau)}\,B\,u(\tau)\,\mathrm{d}\tau, \qquad y(t) = Ce^{At}x_0 + C\int_0^t e^{A(t-\tau)}Bu(\tau)\,\mathrm{d}\tau + Du(t).} \tag{43}$$

This is the *variation of constants* formula. Its first term is the *zero-input response*, what the initial state does on its own; the second is the *zero-state response*. The matrix $e^{A(t-t_0)}$ is the *state transition matrix*: it carries the state at time t_0 to the state at time t when no input acts, $x(t) = e^{A(t-t_0)}x(t_0)$. The integral says the same about the input: the small push $Bu(\tau)\,\mathrm{d}\tau$ delivered at time τ is carried forward to time t by $e^{A(t-\tau)}$, and all the pushes are added up.

Taking Laplace transforms of $\dot{x} = Ax + Bu$ instead gives $X(s) = (sI - A)^{-1}x_0 + (sI - A)^{-1}BU(s)$, which is (43) term by term — a convolution in time is a product in s. Comparing the two,

$$\mathcal{L}\{e^{At}\} = (sI - A)^{-1}. \tag{44}$$

The zero-state part of the output, $C(sI - A)^{-1}BU + DU$, is the transfer function of §5.5 once more; the zero-input part is the free response that the transfer function leaves out.

10.3 Modes: solving by diagonalization

If $A = V\Lambda V^{-1}$ is diagonalizable, change to the coordinates $z = V^{-1}x$ of §9.3. Then $\dot{z} = \Lambda z$ is n separate scalar equations $\dot{z}_i = \lambda_i z_i$, so $z_i(t) = e^{\lambda_i t}z_i(0)$, and going back,

$$x(t) = Vz(t) = z_1(0)\,e^{\lambda_1 t}\,v_1 + \cdots + z_n(0)\,e^{\lambda_n t}\,v_n. \tag{45}$$

Each term is a *mode*: a fixed direction v_i, scaled by a scalar exponential. The eigenvector decides the direction, the eigenvalue decides how fast that component grows or decays, and the initial condition decides how much of each mode is present (Figure 18). The same formula, written without coordinates, uses the *left eigenvectors* w_i^{T} — the rows of V^{-1}, which satisfy $w_i^{\mathsf{T}}A = \lambda_i w_i^{\mathsf{T}}$ and $w_i^{\mathsf{T}}v_j = \delta_{ij}$:

$$e^{At} = Ve^{\Lambda t}V^{-1} = \sum_{i=1}^{n} e^{\lambda_i t}\,v_i w_i^{\mathsf{T}}, \qquad z_i(0) = w_i^{\mathsf{T}}x_0. \tag{46}$$

So e^{At} is a sum of rank-one matrices, one per mode.

Stability of $\dot{x} = Ax$. $x(t) \to 0$ for every initial state if and only if every eigenvalue of A has strictly negative real part.

For diagonalizable A this is visible in (45): a complex pair $\sigma \pm j\omega$ contributes $e^{\sigma t}$ times a rotation at frequency ω, and every term dies out exactly when every $\operatorname{Re}\lambda_i < 0$. §10.4 shows that the statement survives without diagonalizability. It is the state-space twin of the stable-plant condition of Definition 6.1, and since the eigenvalues of A are the poles of P (barring cancellations), the two say the same thing.

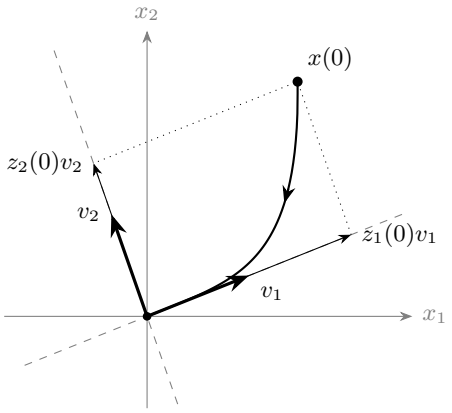


Figure 18. A modal decomposition, drawn for eigenvalues $\lambda_1 = -0.5$ and $\lambda_2 = -2$. The initial state splits along the eigenvectors (dotted parallelogram); each component then shrinks along its own eigenvector at its own rate. The fast mode v_2 dies first, so the trajectory bends and comes in to the origin along the slow direction v_1.

Example 10.2 (one matrix exponential, three ways)**.** Let $A = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}$, with $\det(\lambda I - A) = \lambda^2 + 3\lambda + 2 = (\lambda + 1)(\lambda + 2)$.

By diagonalization. Eigenvectors $v_1 = (1, -1)^{\mathsf{T}}$ for $\lambda_1 = -1$ and $v_2 = (1, -2)^{\mathsf{T}}$ for $\lambda_2 = -2$, so

$$V = \begin{bmatrix} 1 & 1 \\ -1 & -2 \end{bmatrix}, \quad V^{-1} = \begin{bmatrix} 2 & 1 \\ -1 & -1 \end{bmatrix}, \quad e^{At} = e^{-t}\begin{bmatrix} 2 & 1 \\ -2 & -1 \end{bmatrix} + e^{-2t}\begin{bmatrix} -1 & -1 \\ 2 & 2 \end{bmatrix},$$

the two rank-one pieces $v_1 w_1^{\mathsf{T}}$ and $v_2 w_2^{\mathsf{T}}$ of (46).

By Laplace. $(sI - A)^{-1} = \dfrac{1}{(s+1)(s+2)}\begin{bmatrix} s+3 & 1 \\ -2 & s \end{bmatrix}$, and the partial fractions of §5.7 entry by entry, e.g. $\frac{s+3}{(s+1)(s+2)} = \frac{2}{s+1} - \frac{1}{s+2}$, invert to the same matrix.

By Cayley–Hamilton (§10.5). Write $e^{At} = \alpha_0(t)I + \alpha_1(t)A$ and impose $e^{\lambda t} = \alpha_0 + \alpha_1\lambda$ at both eigenvalues: $e^{-t} = \alpha_0 - \alpha_1$, $e^{-2t} = \alpha_0 - 2\alpha_1$. Hence $\alpha_1 = e^{-t} - e^{-2t}$, $\alpha_0 = 2e^{-t} - e^{-2t}$, and $\alpha_0 I + \alpha_1 A$ is once more the matrix above.

10.4 When there are too few eigenvectors

For a matrix in Jordan form the decoupling is incomplete but still easy. With the 3×3 J of §9.5, $\dot{z} = Jz$ reads

$$\dot{z}_1 = \lambda_1 z_1, \qquad \dot{z}_2 = \lambda_2 z_2 + z_3, \qquad \dot{z}_3 = \lambda_2 z_3.$$

The first and last are solved as before. The middle one is driven by z_3, and by (43) with $n = 1$,

$$z_2(t) = e^{\lambda_2 t} z_2(0) + \int_0^t e^{\lambda_2 (t-\tau)} e^{\lambda_2 \tau} z_3(0)\, \mathrm{d}\tau = e^{\lambda_2 t} z_2(0) + t\, e^{\lambda_2 t} z_3(0).$$

Hence

$$x(t) = z_1(0) e^{\lambda_1 t} v_1 + \big(z_2(0) e^{\lambda_2 t} + z_3(0)\, t\, e^{\lambda_2 t} \big) v_2 + z_3(0) e^{\lambda_2 t} v_3,$$

and for a single block $J_2(\lambda)$, $e^{J_2 t} = e^{\lambda t} \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix}$. A Jordan block brings a factor t in front of the exponential — the same t that a repeated pole brought in §5.7. Since $t^k e^{\lambda t} \to 0$ whenever $\operatorname{Re} \lambda < 0$, the stability criterion above holds for every square matrix.

10.5 The Cayley–Hamilton theorem

Theorem 10.3 (Cayley–Hamilton)**.** *Every square matrix satisfies its own characteristic equation: if* $p(\lambda) = \det(\lambda I - A) = \lambda^n + a_{n-1}\lambda^{n-1} + \cdots + a_0$*, then*

$$p(A) = A^n + a_{n-1} A^{n-1} + \cdots + a_1 A + a_0 I = 0.$$

Sketch. If $A = V \Lambda V^{-1}$, then $A^k = V \Lambda^k V^{-1}$, so

$$p(A) = V\, p(\Lambda)\, V^{-1}, \qquad p(\Lambda) = \operatorname{diag}\big(p(\lambda_1), \ldots, p(\lambda_n)\big) = 0,$$

because each λ_i is a root. For a general A the same argument runs block by block on the Jordan form, where p contains a factor $(\lambda - \lambda_i)^k$ large enough to annihilate each $k \times k$ block. □

The consequence is that A^n is a combination of $I, A, \ldots, A^{n-1}$, hence so is $A^{n+1} = A \cdot A^n$, and so on: *every polynomial in* A *reduces to one of degree at most* $n - 1$, and so does every convergent power series — in particular $e^{At} = \alpha_0(t) I + \alpha_1(t) A + \cdots + \alpha_{n-1}(t) A^{n-1}$, as used in Example 10.2. When the eigenvalues are distinct, the coefficients come from the n scalar equations $e^{\lambda_i t} = \sum_k \alpha_k(t) \lambda_i^k$.

Example 10.4 (reducing a matrix polynomial)**.** Let $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, so $p(\lambda) = \lambda^2 - 4\lambda + 3$ and $A^2 = 4A - 3I$. Then $A^3 = 4A^2 - 3A = 13A - 12I$ and $A^4 = 13A^2 - 12A = 40A - 39I$, so

$$A^4 + 3A^3 + A^2 + I = 83A - 77I.$$

By the same token $A^{10000} = aA + bI$ for two numbers a, b; evaluating the scalar identity $\lambda^{10000} = a\lambda + b$ at the eigenvalues 1 and 3 gives $a = \frac{3^{10000} - 1}{2}$, $b = \frac{3 - 3^{10000}}{2}$.

Remark 10.5 (four ways to compute e^{At}). The series definition; the inverse Laplace transform of $(sI - A)^{-1}$ (44); the modal sum $\sum_i e^{\lambda_i t} v_i w_i^{\mathsf{T}}$ when A is diagonalizable (with the Jordan form when it is not); and the Cayley–Hamilton polynomial of degree $n - 1$. They are four views of one object, and Example 10.2 checks three of them against each other.

11 Positive Definiteness and the Singular Value Decomposition

11.1 Quadratic forms and positive definite matrices

A *quadratic form* is $x^{\mathsf{T}} P x$ for a square P. Only the symmetric part of P contributes, since $x^{\mathsf{T}} P x$ is a scalar and so equals its own transpose $x^{\mathsf{T}} P^{\mathsf{T}} x$:

$$x^{\mathsf{T}} P x = x^{\mathsf{T}} \Big(\tfrac{P + P^{\mathsf{T}}}{2} \Big) x.$$

It is therefore no loss to take $P = P^{\mathsf{T}}$.

Definition 11.1 (positive definite)**.** A symmetric P is *positive definite*, written $P \succ 0$, if $x^{\mathsf{T}}Px > 0$ for every $x \neq 0$, and *positive semidefinite*, $P \succeq 0$, if $x^{\mathsf{T}}Px \geq 0$ for every x.

Positive definite matrices are the matrices that measure size: energies, costs and squared distances are all quadratic forms of this kind.

Proposition 11.2 (two tests)**.** *For $P = P^{\mathsf{T}} \in \mathbb{R}^{n\times n}$ the following are equivalent:*

(a) $P \succ 0$;

(b) every eigenvalue of P is positive;

(c) $P = L^{\mathsf{T}}L$ for some L with full column rank.

Proof. By Theorem 9.4, $P = V\Lambda V^{\mathsf{T}}$, so with $y = V^{\mathsf{T}}x$, $x^{\mathsf{T}}Px = y^{\mathsf{T}}\Lambda y = \sum_i \lambda_i y_i^2$. As x ranges over the nonzero vectors so does y, and the sum is positive for all of them exactly when every $\lambda_i > 0$: (a)⇔(b). If $P = L^{\mathsf{T}}L$, then $x^{\mathsf{T}}Px = \|Lx\|^2 \geq 0$, with equality only when $Lx = 0$, i.e. only when $x = 0$ if $\mathcal{N}(L) = \{0\}$: (c)⇒(a). Conversely, (b) lets one take $L = \Lambda^{1/2}V^{\mathsf{T}}$. □

Semidefiniteness is the same with "positive" replaced by "nonnegative" and the rank condition on L dropped. Geometrically, for $P \succ 0$ the level set $x^{\mathsf{T}}Px = 1$ is an ellipsoid whose axes point along the eigenvectors of P, with semi-axis $1/\sqrt{\lambda_i}$ along v_i. For $P = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ the eigenvalues are 1 and 3, so $P \succ 0$, and $2x_1^2 + 2x_1x_2 + 2x_2^2 = 1$ is an ellipse with its long axis along $(1, -1)$.

11.2 The four fundamental subspaces

Every $A \in \mathbb{R}^{m\times n}$ of rank r carries four subspaces, two in each of the spaces it connects:

subspace	lives in	dimension	description
$\mathcal{R}(A^{\mathsf{T}})$	$\mathbb{R}^n$	r	row space
$\mathcal{N}(A)$	$\mathbb{R}^n$	$n - r$	null space
$\mathcal{R}(A)$	$\mathbb{R}^m$	r	column space
$\mathcal{N}(A^{\mathsf{T}})$	$\mathbb{R}^m$	$m - r$	left null space

The pairs are not merely of complementary dimension; they are perpendicular. If $Ax = 0$ and $y = A^{\mathsf{T}}z$, then $y^{\mathsf{T}}x = z^{\mathsf{T}}Ax = 0$, so

$$\mathcal{N}(A) \perp \mathcal{R}(A^{\mathsf{T}}), \qquad \text{and likewise} \qquad \mathcal{N}(A^{\mathsf{T}}) \perp \mathcal{R}(A).$$

Every $x \in \mathbb{R}^n$ thus splits uniquely as $x = x_r + x_n$ with x_r in the row space and x_n in the null space, and $Ax = Ax_r$: the null-space part is lost, and A maps the row space one-to-one onto the column space (Figure 19).

One more fact links A to the symmetric matrix $A^{\mathsf{T}}A$:

$$\mathcal{N}(A^{\mathsf{T}}A) = \mathcal{N}(A), \qquad \text{hence} \qquad \operatorname{rank}(A^{\mathsf{T}}A) = \operatorname{rank} A. \tag{47}$$

Clearly $Ax = 0$ implies $A^{\mathsf{T}}Ax = 0$. Conversely, if $A^{\mathsf{T}}Ax = 0$ then $0 = x^{\mathsf{T}}A^{\mathsf{T}}Ax = \|Ax\|^2$, so $Ax = 0$. In particular $A^{\mathsf{T}}A$ is invertible exactly when A has full column rank — and then $A^{\mathsf{T}}A \succ 0$, by Proposition 11.2(c) with $L = A$. In every case $A^{\mathsf{T}}A \succeq 0$.

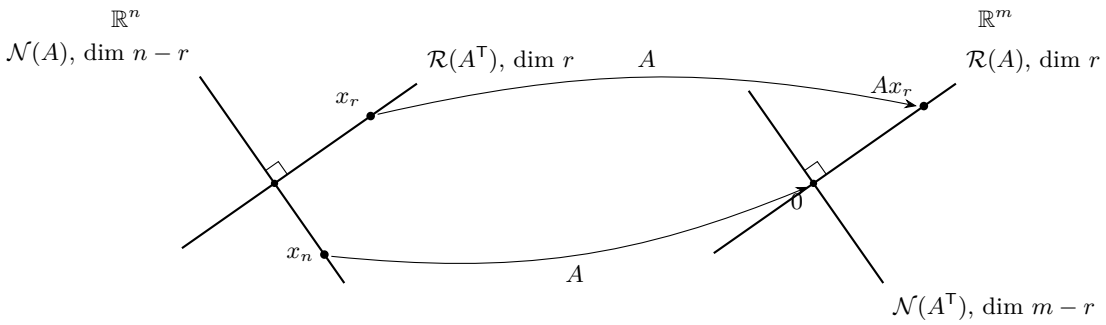


Figure 19. The four fundamental subspaces of $A \in \mathbb{R}^{m\times n}$, each drawn as a line. In $\mathbb{R}^n$ the row space and the null space are perpendicular, and so are the column space and the left null space in $\mathbb{R}^m$. A carries the row-space part x_r of a vector one-to-one onto the column space and sends the null-space part x_n to zero.

11.3 The singular value decomposition

Eigenvectors only make sense for square matrices, and even then they need not be orthogonal or even numerous enough. The singular value decomposition repairs all of this at once, for every matrix.

Theorem 11.3 (SVD)**.** *Every $A \in \mathbb{R}^{m\times n}$ of rank r can be written*

$$A = U\Sigma V^\mathsf{T},$$

where $U \in \mathbb{R}^{m\times m}$ and $V \in \mathbb{R}^{n\times n}$ are orthogonal ($U^\mathsf{T}U = I$, $V^\mathsf{T}V = I$) and $\Sigma \in \mathbb{R}^{m\times n}$ is zero except for the diagonal entries $\sigma_1 \ge \sigma_2 \ge \cdots \ge \sigma_r > 0$. The σ_i are the singular values*, the columns v_i of V the right singular vectors, and the columns u_i of U the left singular vectors. Equivalently,*

$$A = \sum_{i=1}^{r} \sigma_i\, u_i v_i^\mathsf{T}, \qquad Av_i = \sigma_i u_i, \qquad A^\mathsf{T}u_i = \sigma_i v_i. \tag{48}$$

Construction. $A^\mathsf{T}A$ is symmetric and positive semidefinite, so by Theorem 9.4 it has orthonormal eigenvectors $v_1, \dots, v_n$ with eigenvalues $\lambda_1 \ge \cdots \ge \lambda_n \ge 0$, and by (47) exactly r of them are positive. Put $\sigma_i = \sqrt{\lambda_i}$ and $u_i = Av_i/\sigma_i$ for $i \le r$. These are orthonormal:

$$u_i^\mathsf{T}u_j = \frac{v_i^\mathsf{T}A^\mathsf{T}Av_j}{\sigma_i\sigma_j} = \frac{\lambda_j\, v_i^\mathsf{T}v_j}{\sigma_i\sigma_j} = \delta_{ij}.$$

Complete $u_1, \dots, u_r$ to an orthonormal basis of $\mathbb{R}^m$. Then $AV = U\Sigma$, column by column ($Av_i = 0$ for $i > r$, since those v_i lie in $\mathcal{N}(A^\mathsf{T}A) = \mathcal{N}(A)$), and multiplying by V^T on the right gives $A = U\Sigma V^\mathsf{T}$. □

So the singular values are

$$\sigma_i(A) = \sqrt{\lambda_i(A^\mathsf{T}A)},$$

and the singular vectors are orthonormal bases for all four subspaces of §11.2 at once: $v_1, \dots, v_r$ span the row space and $v_{r+1}, \dots, v_n$ the null space; $u_1, \dots, u_r$ span the column space and $u_{r+1}, \dots, u_m$ the left null space. The rank is the number of nonzero singular values. The sum (48) is reading (iii) of §8, $AB = \sum a_i b_i^\mathsf{T}$, with the factors chosen orthonormal and ordered by size.

Example 11.4 (a 2×2 SVD)**.** Let $A = \begin{bmatrix} 3 & 0 \\ 4 & 5 \end{bmatrix}$. Then $A^{\mathsf{T}}A = \begin{bmatrix} 25 & 20 \\ 20 & 25 \end{bmatrix}$, with eigenvalues 45 and 5 and eigenvectors $v_1 = \frac{1}{\sqrt{2}}(1,1)^{\mathsf{T}}$, $v_2 = \frac{1}{\sqrt{2}}(1,-1)^{\mathsf{T}}$. So $\sigma_1 = 3\sqrt{5} \approx 6.71$ and $\sigma_2 = \sqrt{5} \approx 2.24$, and

$$u_1 = \frac{Av_1}{\sigma_1} = \frac{1}{\sqrt{10}} \begin{bmatrix} 1 \\ 3 \end{bmatrix}, \qquad u_2 = \frac{Av_2}{\sigma_2} = \frac{1}{\sqrt{10}} \begin{bmatrix} 3 \\ -1 \end{bmatrix},$$

which are indeed orthogonal. As a check, $\sigma_1 u_1 v_1^{\mathsf{T}} + \sigma_2 u_2 v_2^{\mathsf{T}} = \frac{3}{2} \begin{bmatrix} 1 & 1 \\ 3 & 3 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} 3 & -3 \\ -1 & 1 \end{bmatrix} = A$, and $\sigma_1 \sigma_2 = 15 = |\det A|$. The eigenvalues of A are 3 and 5 — not the singular values. For a non-symmetric matrix the two are different objects; for a symmetric one the singular values are the absolute values of the eigenvalues.

11.4 What the singular values measure

Read $A = U\Sigma V^{\mathsf{T}}$ from right to left: V^{T} rotates the input (it is orthogonal, so it preserves lengths), Σ stretches the i-th coordinate by σ_i, and U rotates the result. Hence A maps the unit sphere onto an ellipsoid whose semi-axes are $\sigma_i u_i$, and the input directions that end up on those axes are the v_i (Figure 20).

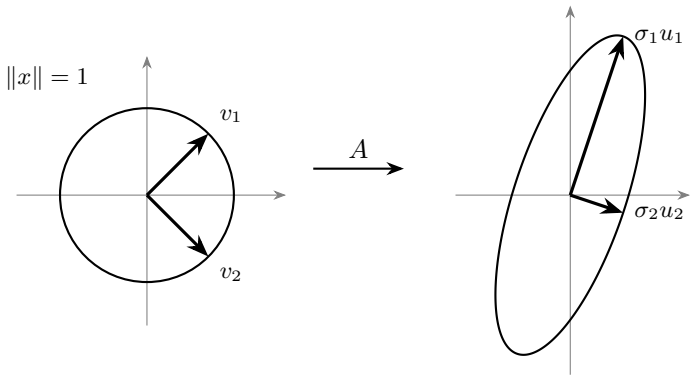


Figure 20. The SVD of Example 11.4 as a picture. The unit circle (left) is carried by A onto an ellipse (right, drawn at one third the scale) with semi-axes $\sigma_1 u_1$ and $\sigma_2 u_2$, of lengths $3\sqrt{5}$ and $\sqrt{5}$. The input directions v_1 and v_2 are the ones that land on the axes.

Several useful numbers follow directly.

- **Gain.** For every $x \neq 0$, $\sigma_{\min} \leq \|Ax\|/\|x\| \leq \sigma_1$ (with $\sigma_{\min} = \sigma_n$ if A has full column rank, and 0 otherwise). The largest amplification is σ_1, attained for x along v_1; it is the *induced* 2*-norm* $\|A\|_2 = \sigma_1$. So singular values say how much a matrix can amplify, and in which input direction — where eigenvalues say how fast each mode evolves.
- **Conditioning.** The ratio $\kappa(A) = \sigma_1/\sigma_r$ is the *condition number*. When it is large, some input directions are amplified far more than others, and solving $Ax = b$ magnifies errors in b by up to κ. A matrix that is invertible on paper but has σ_r tiny is, numerically, nearly rank-deficient.
- **Low-rank approximation.** Truncating (48) after k terms, $A_k = \sum_{i \leq k} \sigma_i u_i v_i^{\mathsf{T}}$, gives the closest matrix of rank k to A, with error $\|A - A_k\|_2 = \sigma_{k+1}$. The singular values say exactly how much is lost by keeping only the strongest directions.

Remark 11.5 (least squares). When $Ax = b$ has no solution ($b \notin \mathcal{R}(A)$), the SVD still gives the best one can do. The *pseudo-inverse* $A^{+} = V\Sigma^{+}U^{\mathsf{T}}$, where Σ^{+} inverts the nonzero singular values

and transposes the shape, yields $x = A^{+}b$, which minimizes $\|Ax - b\|$ and, among all minimizers, has the smallest norm. If A has full column rank, $A^{+} = (A^{\mathsf{T}}A)^{-1}A^{\mathsf{T}}$, which is invertible by (47).